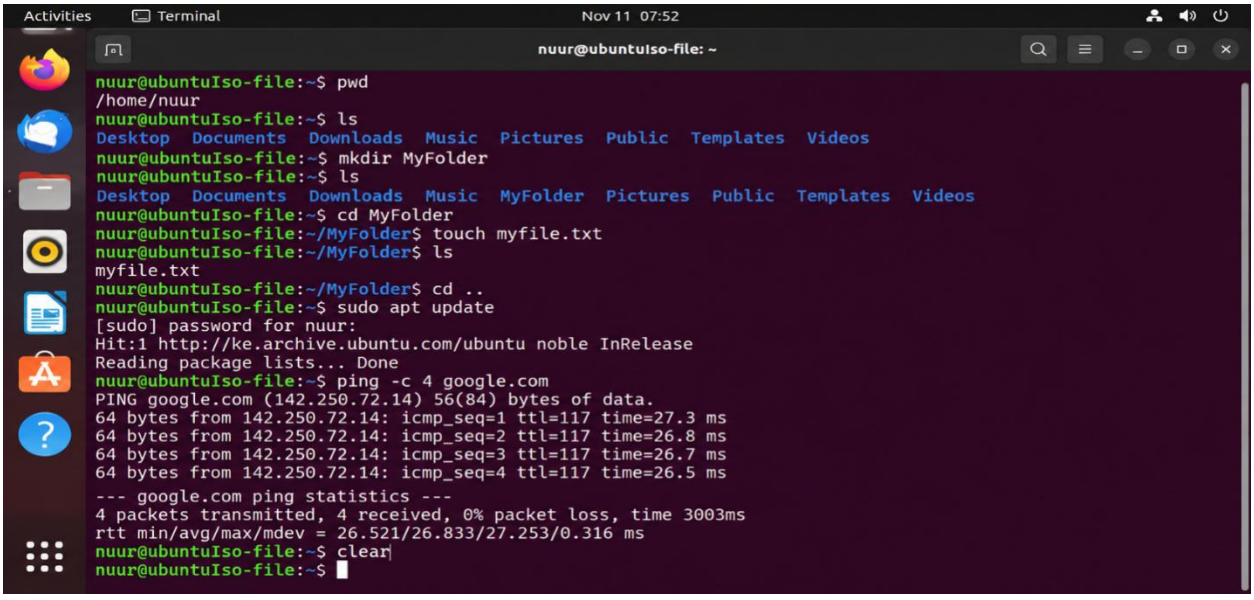
- ❖ Linux is an open-source operating system (OS) that manages computer hardware and software. It is similar to Windows and macOS.
- ❖ Linux allows you to:
  - Run applications and programs , Manage files and folders ,Connect to networks and the internet, Manage servers, Write and execute programming code And Configure network security.

## Basic Linux commands

| Command | Meaning |
|---|---|
| `pwd` | Show current directory |
| `ls` | List files |
| `cd` | Change directory |
| `mkdir` | Create a folder |
| `touch` | Create a file |
| `clear` | Clear terminal |
| `sudo` | Run as administrator |
| `ping` | Test network connection |



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# What is `/proc` in Linux?

`/proc` is a special directory in Linux that contains information about the system, hardware, memory, CPU, and running processes.

It is called a virtual filesystem because most of its information is created by the Linux kernel. It does not work like a normal folder that stores ordinary files.

The `/proc` directory helps users check system information using terminal commands.

Location: `/proc`

## 1. Common files and directories inside `/proc`

| File/Directory | |
|---|---|
| **`/proc/cpuinfo=`** | Shows CPU information. |
| **`/proc/meminfo=`** | Shows RAM and memory information. |
| **`/proc/uptime=`** | Shows how long the system has been running. |
| **`/proc/version=`** | Shows Linux kernel version. |
| **`/proc/mounts=`** | Shows mounted filesystems. |
| **`/proc/cmdline =`** | Shows kernel startup parameters. |
| **`/proc/loadavg =`** | Shows system load averages. |
| **`/proc/PID/ =`** | Contains information about a running process. |

## Options and commands used with `/proc`

Important: `/proc` itself does not have command options. We use Linux commands such as `ls`, `cat`, and `grep` to read information from it.

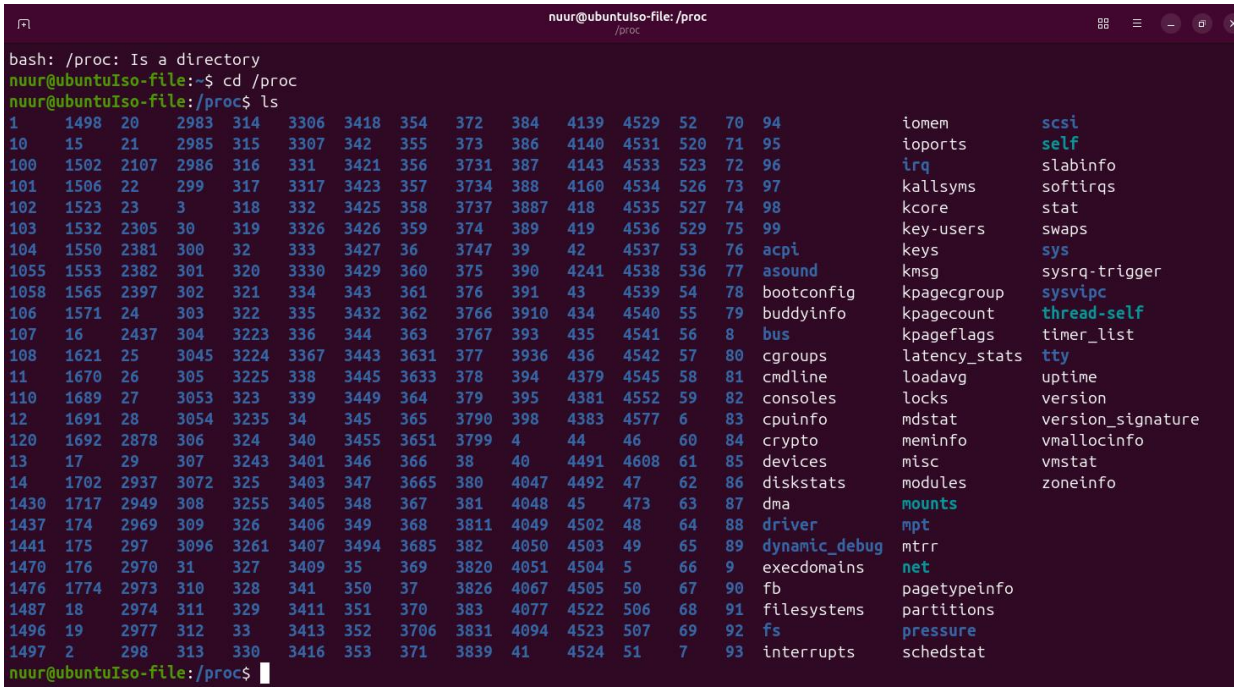
A. `ls` – List files

Shows files and directories inside `/proc`.

Options:

- `-l`: Shows detailed information.
- `-a`: Shows hidden entries.
- `-h`: Displays file sizes in readable format when used with `-l`.

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## What is `/sys` in Linux?

- `/sys` is a special directory in Linux that contains information about hardware devices, drivers, and kernel components.

It is called a virtual filesystem (sysfs). It allows users to view information about devices connected to the computer and how Linux manages them.

| Directory | Function |
|---|---|
| **`/sys/class`** | = Shows devices grouped by type. |

**`/sys/block`**=Shows block devices such as hard disks.

**`/sys/bus`**=Shows devices and drivers connected through system buses.

**`/sys/devices`**=Contains information about hardware devices.

**`/sys/module`**=Shows loaded kernel modules.

**`/sys/firmware`**=Contains information about system firmware.

**`/sys/kernel`**=Shows information about the Linux kernel

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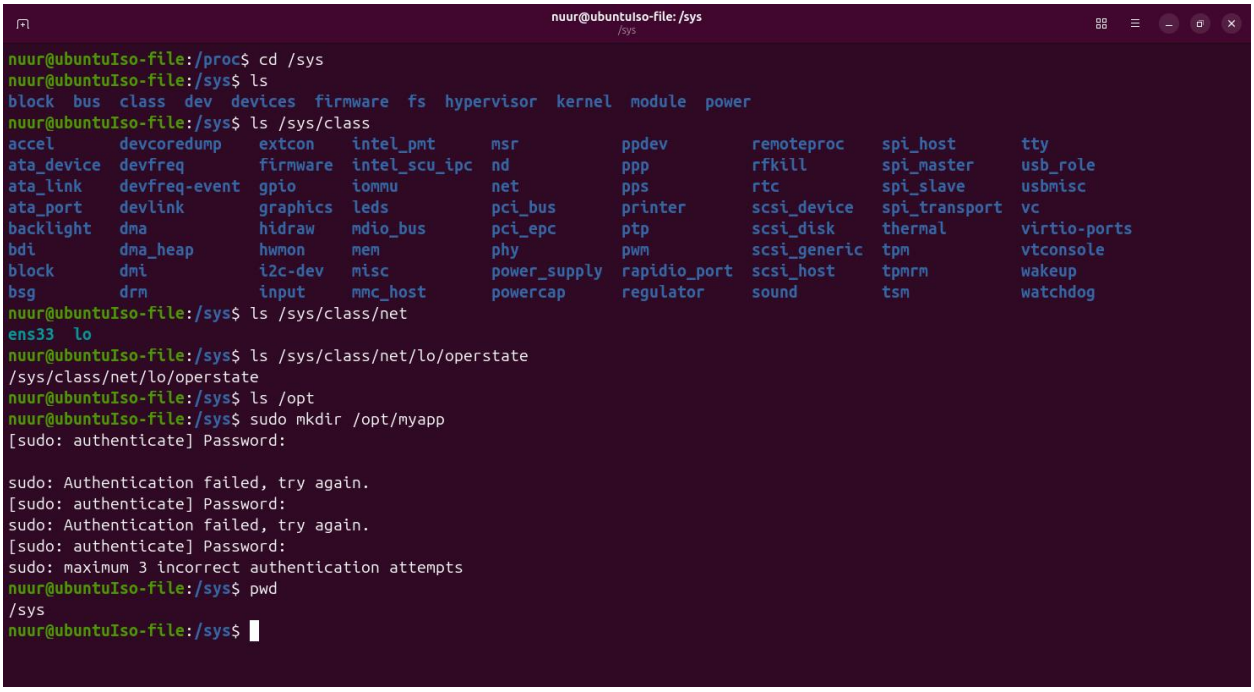
## 2. Commands and options used with `/sys`

Important: `/sys` does not have its own command options. We use Linux commands to access its information.

A. `ls` – List devices and directories

Options:

- `-l`: Shows detailed information.
- `-a`: Shows hidden entries.
- `-R`: Lists directories and their contents recursively.

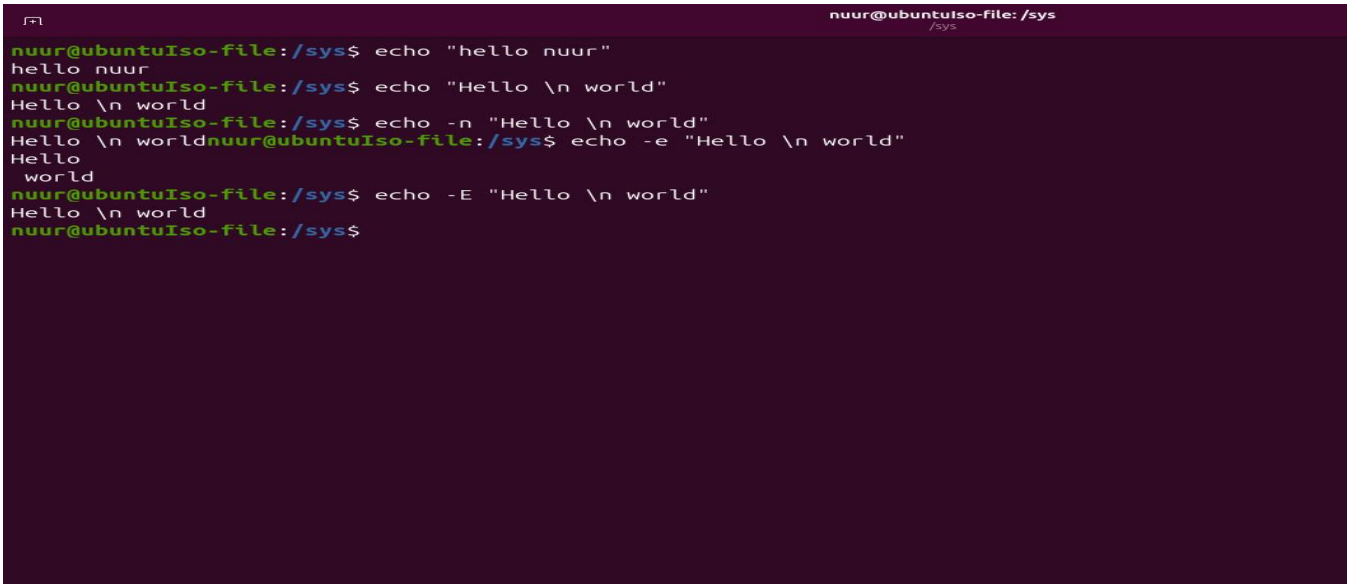



**echo** is a Linux command used to display text or variable values in the terminal.

Common options:

- `-n` – Does not print a new line at the end.
- `-e` – Enables special characters such as `\n` and `\t`.
- `-E` – Disables interpretation of special characters (default in many shells).

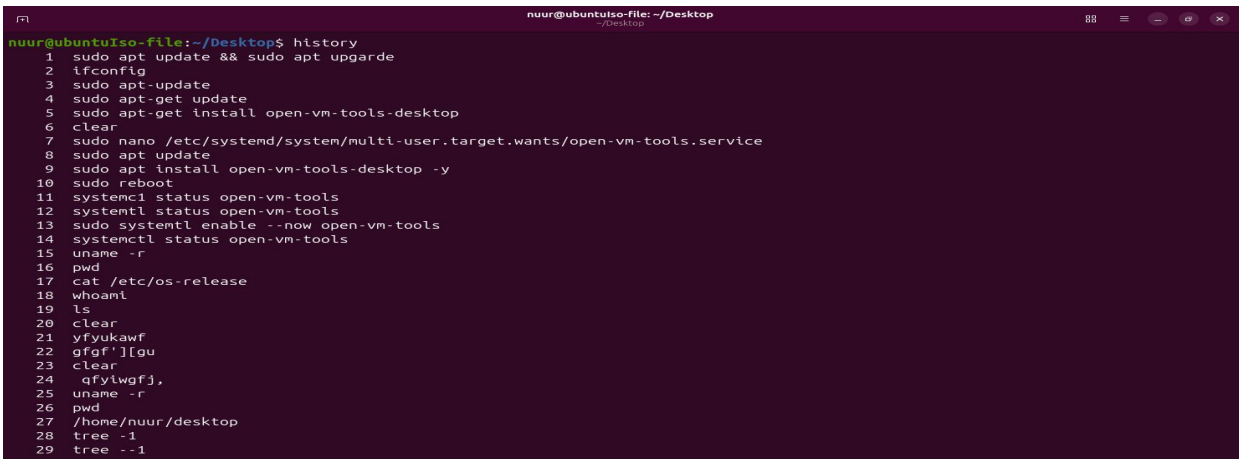
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**The `history`** command in Linux is used to display a list of commands that you have previously executed in the terminal.

**Function:** Displays previously executed commands with their numbers.

| Command | Function |
|---|---|
| `history 5` | Shows the last 5 commands |
| `history -c` | Clears current history |
| `history -d 5` | Deletes entry number 5 |
| `history -w` | Saves history |
| `history -r` | Reloads history |
| `!10` | Executes command number 10 |
| `history \| grep ls` | Searches for `ls` commands |



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The **cd** command stands for Change Directory. It is used in Linux to move from one directory (folder) to another directory in the terminal.

Function of cd:

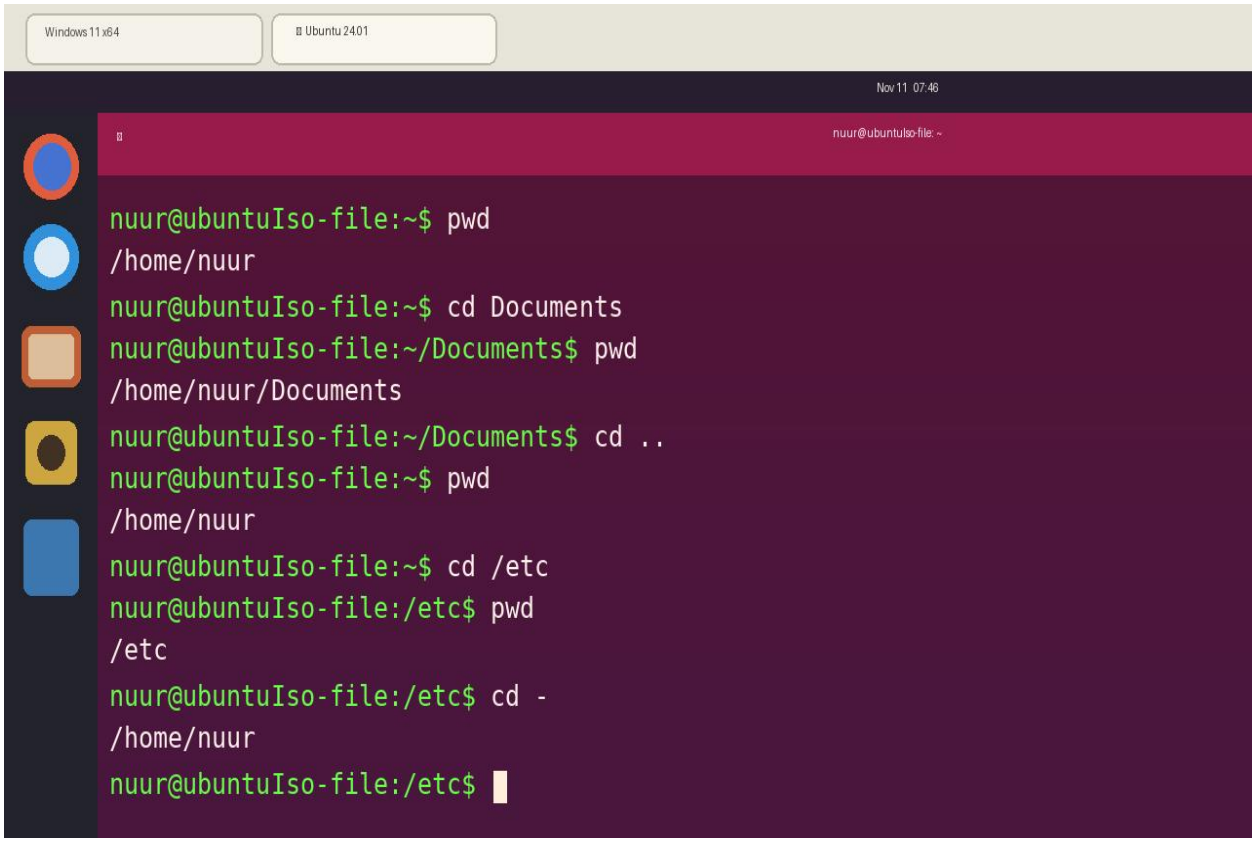
The main function of `cd` is to change the current working directory. It helps users navigate between folders in the Linux filesystem.

## Syntax:

**`cd [OPTION] [DIRECTORY]`**

Common options and uses of cd

1. **`cd`** – Go to Home Directory , Function: Returns the user to their home directory.
2. **`cd ~`** – Go to Home Directory, Function: Moves to the user's home directory.
3. **`cd ..`** – Go to Parent Directory , Function: Moves one level up from the current directory.
4. **`cd /`** – Go to Root Directory, Function: Moves to the root directory of Linux.
5. **`cd -`** – Return to Previous Directory, Function: Returns to the previous working directory.
6. **`cd /etc`** – Go to a Specific Directory, Function: Moves directly to the `/etc` directory.



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## `ls` Command and Its Options

### Definition

The **`ls`** command is used to list files and directories in the current directory. It helps users see the contents of a folder.

Students use **`ls`** to check files and folders, find hidden files, and view file details such as permissions, owner, size, and date.

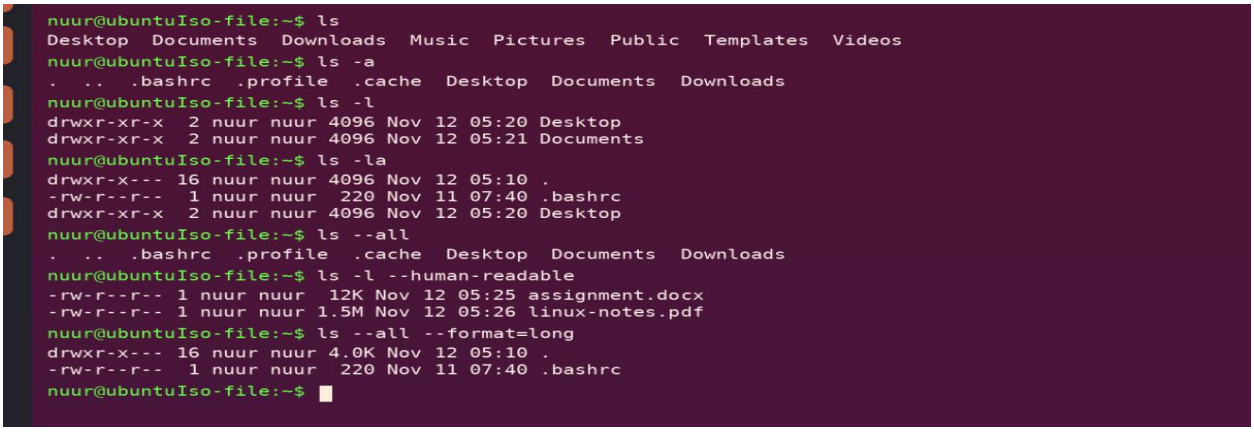
### Syntax

```
ls [option] [directory]
```

### Options and shortcuts

| Option / Command | Name | Description and example usage |
|---|---|---|
| `ls` | List files | Shows normal files and directories. Example: `ls` |
| `ls -a` | Show all | Shows hidden files too. Example: `ls -a` |
| `ls -l` | Long listing | Shows permissions, owner, size, and date. Example: `ls -l` |
| `ls -la` | Long listing + all | Shows hidden files with detailed information. Example: `ls -la` |
| `ls --all` | Show all | Long version of `-a`. Example: `ls --all` |
| `ls --human-readable` | Human-readable sizes | Displays file sizes in an easier format when used with a size listing option such as `-l`. Example: `ls -l --human-readable` |
| `ls --all --format=long` | All files in long format | Shows hidden files and detailed file information. Example: `ls --all --format=long` |

Note: `ls --human-readable` is more useful with `-l`, for example `ls -lh`, because the long listing displays file sizes.




## `man` Command

### Definition

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The **man** command means Manual. It is used to display the manual pages of Linux commands. It helps users understand how a command works, its options, and how to use it.

Students use `man` to learn about Linux commands and understand their options. It is useful when they do not know how a command works.

### Syntax

```
man [option] command
```

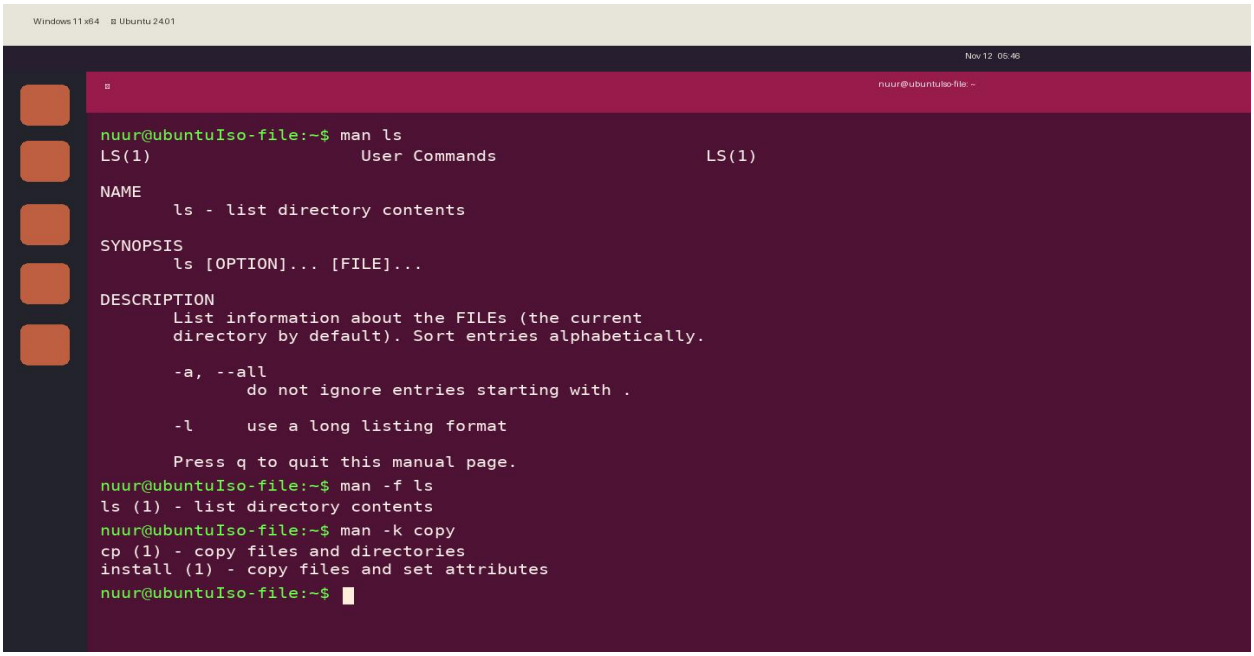
### Options and shortcuts

| Option / Command | Name | Description and example usage |
|---|---|---|
| `man ls` | Manual page | Displays the manual for the `ls` command. |
| `man pwd` | Manual page | Displays information about the `pwd` command. |
| `man -f ls` | Short description | Shows a short description of the `ls` command. |
| `man -k copy` | Search manual | Searches manual pages related to the word `copy`. |
| `man -a passwd` | Show all pages | Displays all matching manual pages for `passwd`. |
| `man 5 passwd` | Manual section | Opens section 5 of the `passwd` manual. |
| `man -w ls` | Page location | Shows the location of the manual page for `ls`. |

### Useful keys inside a manual page

| Key | Function |
|---|---|
| `Space` | Move down one page. |
| `b` | Move up one page. |
| `/word` | Search for a word. |
| `n` | Find the next match. |
| `q` | Quit the manual page. |

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## help and info Commands

### DEFINITION

The **`help`** and **`info`** commands are used to learn about Linux commands and understand how they work.

- 

help: Displays help information about Bash built-in commands, such as `cd`, `echo`, and `history`.

- 
- 

info: Opens detailed documentation about Linux programs and their features.

- 

Students use `help` and `info` to learn command syntax, understand available options, and find instructions for using Linux commands.

### SYNTAX

```
help [OPTION] [COMMAND]
info [OPTION] [MENU-ITEM]
```

### OPTIONS AND SHORTCUTS

help command

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| Option | Name | Description & Example Usage |
|---|---|---|
| `help` | List help | Displays a list of Bash built-in commands. Example: `help` |
| `help cd` | Command help | Shows help information about `cd`. |
| `help echo` | Command help | Shows help information about `echo`. |
| `help -d cd` | Short description | Displays a short description of `cd`. |
| `help -s cd` | Syntax | Displays the syntax of `cd`. |
| `help -m cd` | Manual format | Displays help in a manual-style format. |

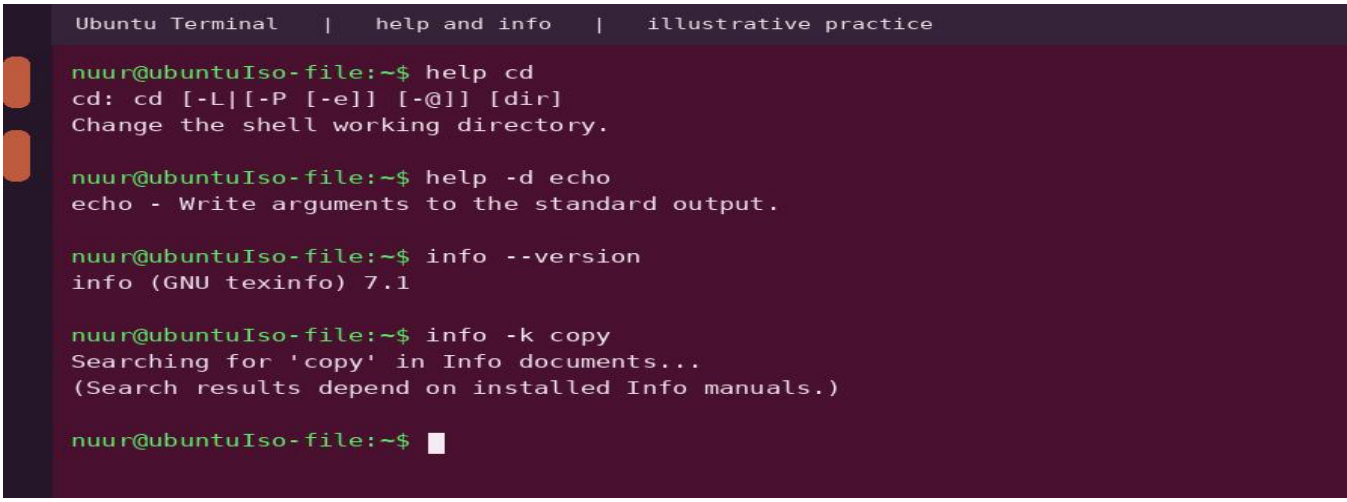
info command

| Option | Name | Description & Example Usage |
|---|---|---|
| `info` | Open Info | Opens the Info documentation menu. |
| `info ls` | Command information | Opens documentation about `ls`. |
| `info -k copy` | Search information | Searches Info documents for the word `copy`. |
| `info -f coreutils` | Open manual file | Opens the Coreutils documentation. |
| `info -w ls` | Show file location | Displays the location of the Info document for `ls`. |

## USEFUL KEYS IN INFO

| Key | Function |
|---|---|
| `Space` | Move down. |
| `Backspace` | Move up. |
| `n` | Go to the next node. |
| `p` | Go to the previous node. |
| `u` | Go to the upper-level node. |
| `q` | Quit Info. |

---




## touch Command

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## DEFINITION

The **touch** command is used to create an empty file in Linux. It can also update the access time and modification time of an existing file without changing its contents.

For example, `touch notes.txt` creates a new empty file named `notes.txt` if it does not already exist.

Students use the `touch` command to create files quickly, prepare files for assignments, and update file timestamps.

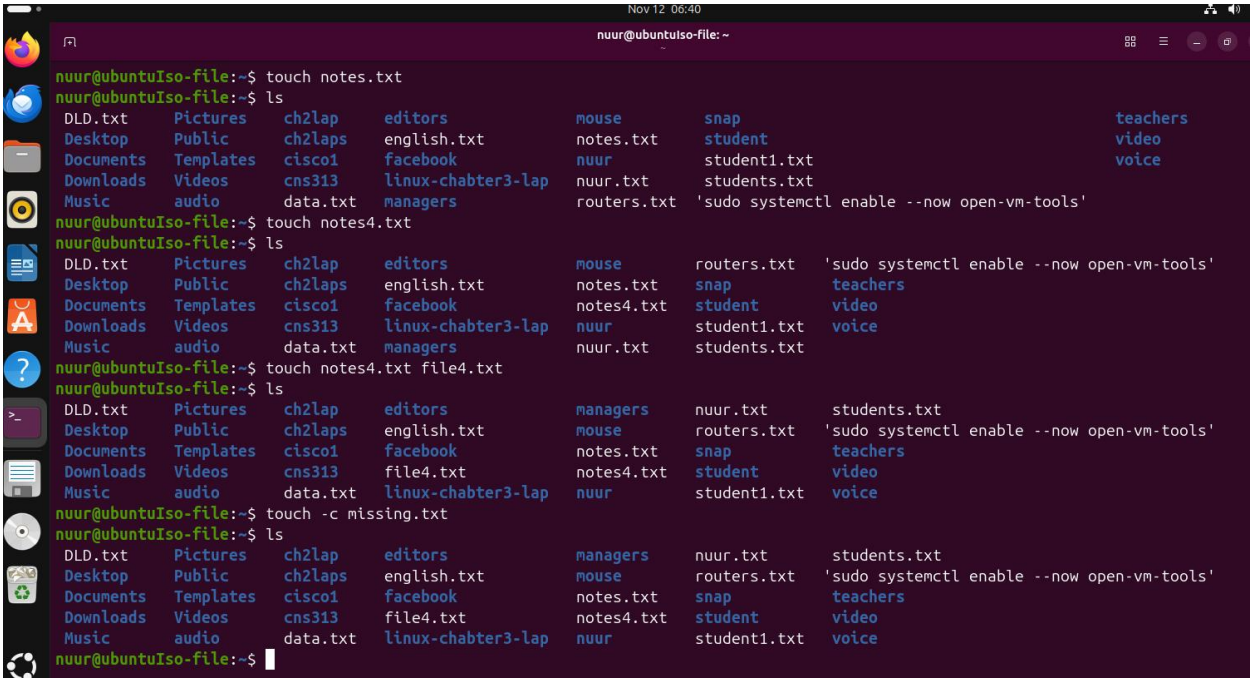
## SYNTAX

```
touch [OPTION]... FILE...
```

## OPTIONS AND SHORTCUTS

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-a` | Access time | Updates the access time. Example: `touch -a notes.txt` |
| `-m` | Modification time | Updates the modification time. Example: `touch -m notes.txt` |
| `-c` | No create | Does not create a file if it does not exist. Example: `touch -c test.txt` |
| `-d` | Date string | Sets the time using a date string. Example: `touch -d "2026-01-01" notes.txt` |
| `-r` | Reference file | Copies timestamps from another file. Example: `touch -r old.txt new.txt` |
| `-t` | Timestamp | Sets a specific timestamp. Example: `touch -t 202601011200 notes.txt` |
| `--help` | Help | Displays help information. Example: `touch --help` |
| `--version` | Version | Displays the touch version. Example: `touch --version` |

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## Linux Path Symbols: ., .., and ~

### DEFINITION

In Linux, ., .., and ~ are special symbols used to represent directory locations.

- 

. (Single dot): Represents the current directory.

- 
- 

.. (Double dot): Represents the parent directory (one level above the current directory).

- 
- 

~ (Tilde): Represents the home directory of the current user.

Students use these symbols to navigate between directories easily without typing full directory paths. They make working with files and folders faster.

### SYNTAX

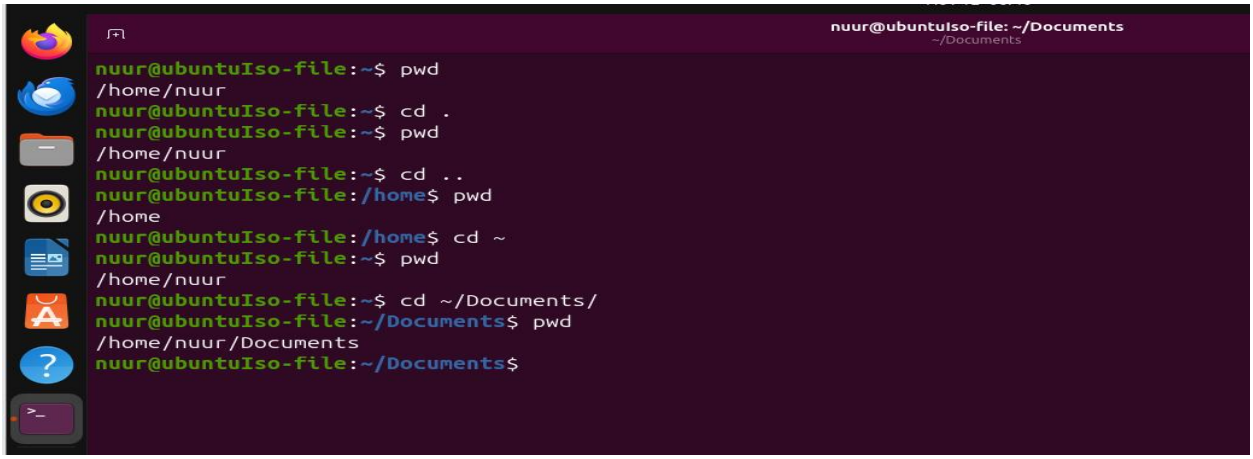
```
cd .
cd ..
cd ~
```

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## OPTIONS AND SHORTCUTS

| Symbol | Name | Description & Example Usage |
|---|---|---|
| `.` | Current directory | Refers to the directory you are currently in. Example: `cd .` |
| `..` | Parent directory | Moves one level up. Example: `cd ..` |
| `~` | Home directory | Takes the user to their home directory. Example: `cd ~` |
| `./` | Current directory path | Accesses a file in the current directory. Example: `./notes.txt` |
| `../` | Parent directory path | Accesses a file in the parent directory. Example: `cd ../Documents` |
| `~/` | Home directory path | Accesses a folder inside the home directory. Example: `cd ~/Documents` |

Important: `.` and `..` are directory symbols, not commands. `~` represents the current user's home directory.




## mkdir and rmdir Commands

### 1. mkdir Command

DEFINITION

The **`mkdir`** command means Make Directory. It is used to create new directories (folders) in Linux.

For example, `mkdir Documents` creates a new directory called `Documents`.

WHY STUDENTS USE THIS COMMAND

Students use `mkdir` to organize their assignment files, create folders for different subjects, and keep their work organized.

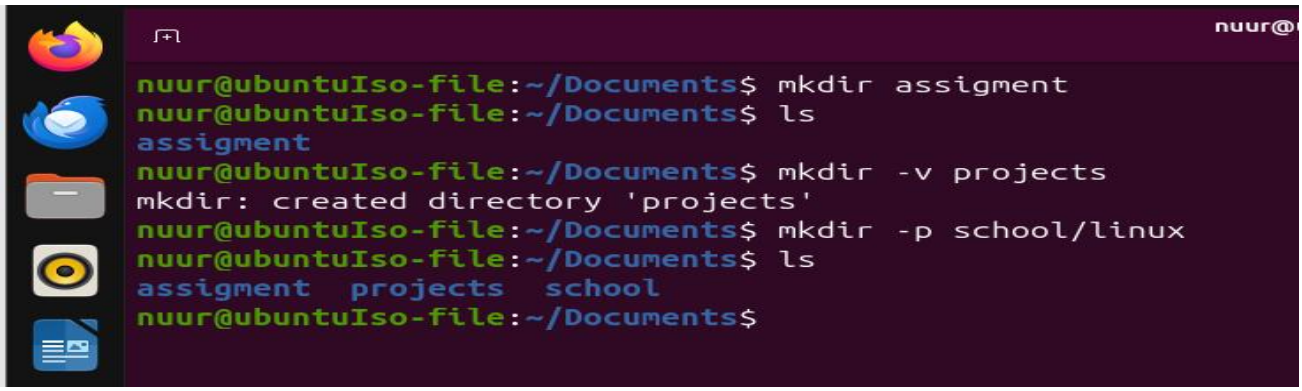
SYNTAX

Linux Assigment name= Nuur abdullahi mohamud ID=c6240109

```
mkdir [OPTION]... DIRECTORY...
```

OPTIONS AND SHORTCUTS

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-p` | Parent directories | Creates parent folders if they do not exist. Example: `mkdir -p school/linux` |
| `-m` | Mode | Sets directory permissions. Example: `mkdir -m 755 projects` |
| `-v` | Verbose | Displays a message when creating a directory. Example: `mkdir -v files` |
| `--parents` | Create parents | Long version of `-p`. Example: `mkdir --parents school/linux` |
| `--verbose` | Verbose mode | Long version of `-v`. Example: `mkdir --verbose projects` |
| `--help` | Help | Displays command help. Example: `mkdir --help` |




## rmdir Command

DEFINITION

The `rmdir` command means Remove Directory. It is used to delete empty directories in Linux. It cannot normally remove a directory that contains files or other folders.

For example, `rmdir Test` removes the empty directory named `Test`.

Students use `rmdir` to remove empty folders that are no longer needed and keep their directories organized.

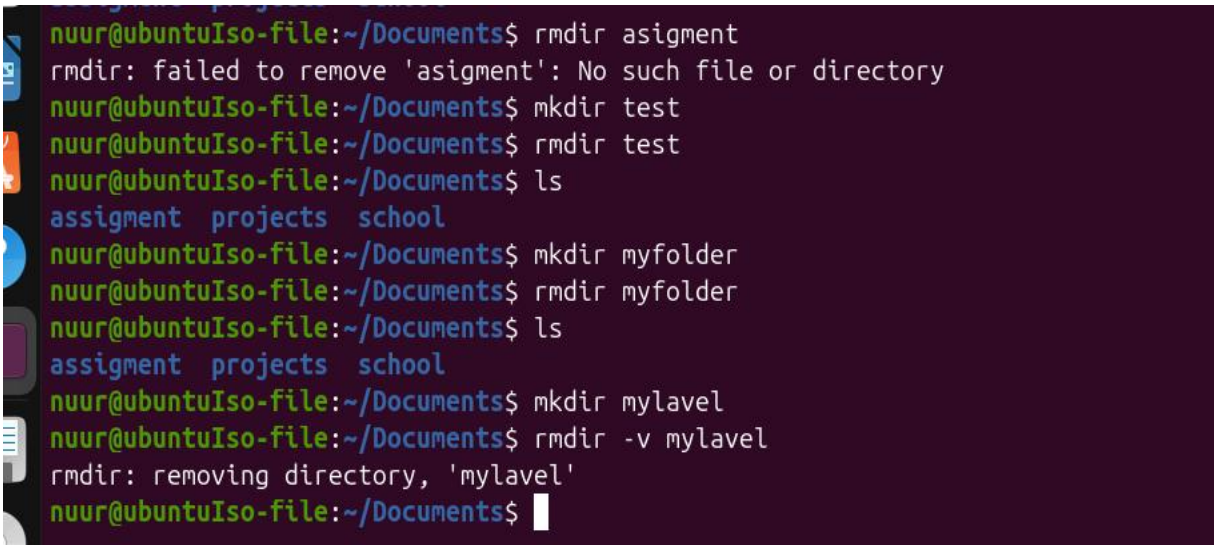
SYNTAX

```
rmdir [OPTION]... DIRECTORY...
```

OPTIONS AND SHORTCUTS

Linux Assigment name= Nuur abdullahi mohamud ID=c6240109

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-p` | Parent directories | Removes empty parent directories too. Example: `rmdir -p School/Linux` |
| `-v` | Verbose | Displays a message when removing a directory. Example: `rmdir -v Test` |
| `--ignore-fail-on-non-empty` | Ignore non-empty errors | Does not report an error for directories that are not empty. Example: `rmdir --ignore-fail-on-non-empty Test` |
| `--parents` | Remove parents | Long version of `-p`. Example: `rmdir --parents School/Linux` |
| `--verbose` | Verbose mode | Long version of `-v`. Example: `rmdir --verbose Test` |
| `--help` | Help | Displays command help. Example: `rmdir --help` |




## cp and mv Commands

### 1. cp Command

#### DEFINITION

The `cp` command means Copy. It is used to copy files and directories from one location to another without removing the original file.

For example, `cp notes.txt backup.txt` creates a copy of `notes.txt` named `backup.txt`.

Students use the `cp` command to make backup copies of their assignment files, copy documents between folders, and keep extra copies of important files.

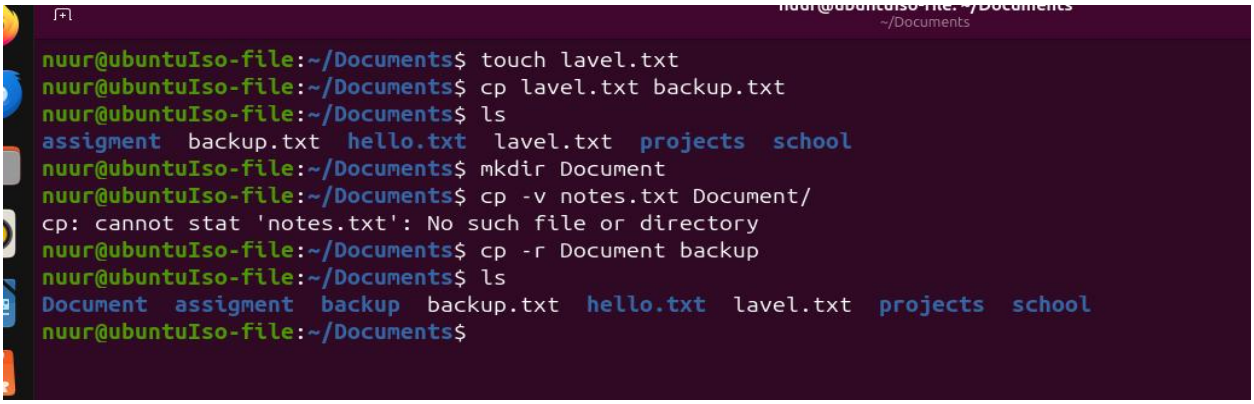
#### SYNTAX

```
cp [OPTION]... SOURCE DESTINATION
```

#### OPTIONS AND SHORTCUTS

Linux Assigment name= Nuur abdullahi mohamud ID=c6240109

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-r` | Recursive | Copies directories and their contents. Example: `cp -r Folder Backup` |
| `-i` | Interactive | Asks before overwriting an existing file. Example: `cp -i notes.txt backup.txt` |
| `-v` | Verbose | Displays a message showing copied files. Example: `cp -v notes.txt Documents/` |
| `-u` | Update | Copies only when the source is newer or the destination is missing. Example: `cp -u notes.txt Backup/` |
| `-n` | No clobber | Does not overwrite an existing destination file. Example: `cp -n notes.txt backup.txt` |
| `-p` | Preserve | Keeps file permissions and timestamps. Example: `cp -p notes.txt Backup/` |
| `-a` | Archive | Copies directories recursively and preserves file attributes. Example: `cp -a Folder Backup/` |
| `--help` | Help | Displays command help. Example: `cp --help` |




## mv Command

### DEFINITION

The `mv` command means Move. It is used to move files and directories from one location to another. It can also rename files and directories.

For example, `mv notes.txt newnotes.txt` changes the name of `notes.txt` to `newnotes.txt`.

Students use the `mv` command to organize files, move assignments into different folders, and rename documents.

### SYNTAX

```
mv [OPTION]... SOURCE DESTINATION
```

### OPTIONS AND SHORTCUTS

Linux Assigment name= Nuur abdullahi mohamud ID=c6240109

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-i` | Interactive | Asks before overwriting an existing file. Example: `mv -i notes.txt Documents/` |
| `-f` | Force | Moves without asking before overwriting. Example: `mv -f notes.txt Documents/` |
| `-n` | No clobber | Does not overwrite an existing destination file. Example: `mv -n notes.txt Documents/` |
| `-u` | Update | Moves only when the source is newer or the destination is missing. Example: `mv -u notes.txt Documents/` |
| `-v` | Verbose | Displays a message showing moved files. Example: `mv -v notes.txt Documents/` |
| `--help` | Help | Displays command help. Example: `mv --help` |




## printf Command

### DEFINITION

The `printf` command means Print Formatted. It is used to display text and values in a specific format in the Linux terminal.

It is similar to the `echo` command, but `printf` gives users more control over spacing, numbers, and formatting.

For example, `printf "Hello World\n"` displays the text Hello World and moves to a new line.

Students use the `printf` command to display information neatly, format text, print numbers, and create organized output in the terminal.

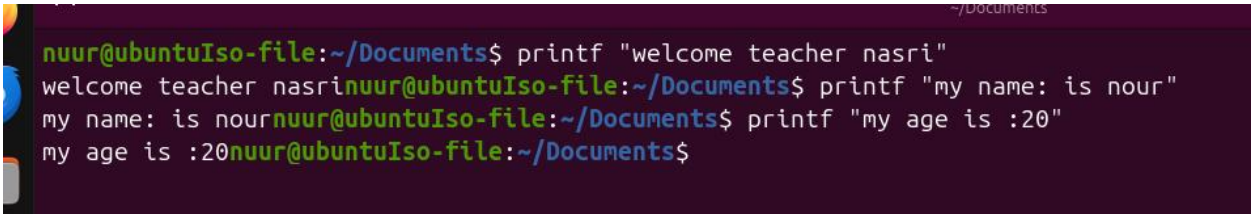
### SYNTAX

```
printf FORMAT [ARGUMENT]...
```

### OPTIONS AND SHORTCUTS

| Option / Format | Name | Description & Example Usage |
|---|---|---|

| | | |
|---|---|---|
| `%s` | String | Displays text. Example: `printf "%s" "Hello"` |
| `%d` | Integer | Displays a whole number. Example: `printf "%d" 25` |
| `%f` | Decimal | Displays a decimal number. Example: `printf "%f" 3.14` |
| `\n` | New line | Moves output to the next line. Example: `printf "Hello\nWorld\n"` |
| `\t` | Tab | Adds a tab space. Example: `printf "Name\tAge\n"` |
| `%%` | Percent sign | Displays a percent symbol. Example: `printf "Success: 90%%\n"` |
| `%-10s` | Left alignment | Aligns text to the left in a 10-character space. Example: `printf "%-10s" "Linux"` |
| `%10s` | Right alignment | Aligns text to the right in a 10-character space. Example: `printf "%10s" "Linux"` |




## cat Command

### DEFINITION

The `cat` command means Concatenate. It is used to display the contents of files in the Linux terminal. It can also create files, combine files, and add text to files.

For example, `cat notes.txt` displays the contents of a file named `notes.txt`.

Students use the `cat` command to read text files, check assignment contents, create simple text files, and combine information from different files.

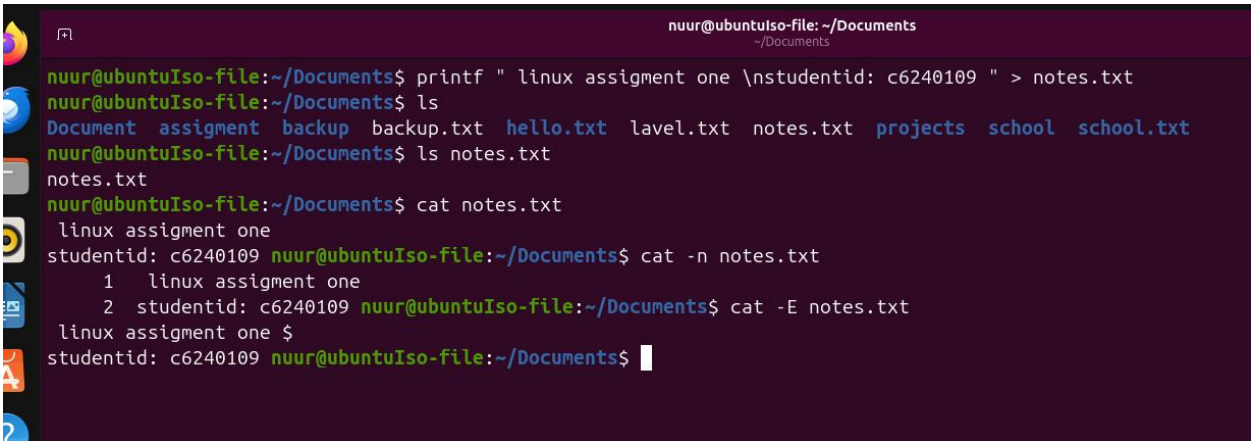
### SYNTAX

```
cat [OPTION]... [FILE]...
```

### OPTIONS AND SHORTCUTS

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-n` | Number all lines | Displays line numbers for all lines. Example: `cat -n notes.txt` |
| `-b` | Number non-empty lines | Numbers only lines that contain text. Example: `cat -b notes.txt` |
| `-s` | Squeeze blank lines | Reduces repeated empty lines to one. Example: `cat -s notes.txt` |

| | | |
|---|---|---|
| `-A` | Show all characters | Displays line endings and tab characters. Example: `cat -A notes.txt` |
| `-E` | Show line endings | Displays `$` at the end of each line. Example: `cat -E notes.txt` |
| `-T` | Show tabs | Displays tab characters as `^I`. Example: `cat -T notes.txt` |
| `-v` | Show special characters | Displays non-printing characters in a visible form. Example: `cat -v notes.txt` |
| `--help` | Help | Displays command help. Example: `cat --help` |




## less and more Commands

### 1. less Command

#### DEFINITION

The `less` command is used to view the contents of a text file one screen at a time. It allows users to move forward and backward through a file without changing its contents.

For example, `less notes.txt` opens the file and displays its contents page by page.

Students use the `less` command to read long text files, view large documents, search for specific words, and move through file contents easily.

#### SYNTAX

```
less [OPTION]... FILE
```

#### OPTIONS AND SHORTCUTS

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-N` | Line numbers | Displays line numbers. Example: `less -N notes.txt` |
| `-S` | Chop long lines | Prevents long lines from wrapping. Example: `less -S notes.txt` |

| | | |
|---|---|---|
| `-i` | Ignore case | Searches without considering uppercase or lowercase. Example: `less -i notes.txt` |
| `-F` | Quit if one screen | Automatically exits if the file fits on one screen. Example: `less -F notes.txt` |
| `-X` | No screen clearing | Keeps the displayed text on the terminal after exiting. Example: `less -X notes.txt` |
| `/word` | Search forward | Searches for a word inside the file. Example: `/Linux` |
| `n` | Next match | Finds the next search result. |
| `q` | Quit | Closes the file viewer. |

### USEFUL KEYS IN less

| Key | Function |
|---|---|
| `Space` | Moves down one screen. |
| `b` | Moves up one screen. |
| `Down Arrow` | Moves down one line. |
| `Up Arrow` | Moves up one line. |
| `g` | Goes to the beginning of the file. |
| `G` | Goes to the end of the file. |
| `/word` | Searches for a word. |
| `n` | Moves to the next match. |
| `q` | Exits the viewer. |

## more Command

### DEFINITION

The `more` command is used to display the contents of a text file one screen at a time. It is mainly used to read long files when the contents cannot fit on one screen.

For example, `more notes.txt` displays the file contents page by page.

Students use the `more` command to read long files, view information page by page, and check text without opening a text editor.

### SYNTAX

```
more [OPTION]... FILE
```

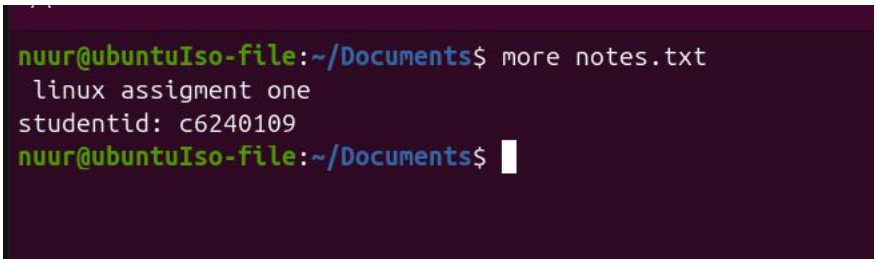
### OPTIONS AND SHORTCUTS

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-d` | Display help prompt | Shows helpful messages when moving through a file. Example: `more -d notes.txt` |

| | | |
|---|---|---|
| `-s` | Squeeze blank lines | Reduces repeated blank lines to one. Example: `more -s notes.txt` |
| `-f` | Count logical lines | Counts logical lines instead of screen lines. Example: `more -f notes.txt` |
| `-p` | Clear and display | Clears the screen before displaying each page. Example: `more -p notes.txt` |
| `-c` | Clear without scrolling | Displays each page without scrolling the previous page. Example: `more -c notes.txt` |
| `-n` | Set screen lines | Sets the number of lines displayed per screen. Example: `more -n 10 notes.txt` |
| `+number` | Start at line | Starts displaying from a specified line. Example: `more +5 notes.txt` |

### USEFUL KEYS IN more

| Key | Function |
|---|---|
| `Space` | Moves to the next screen. |
| `Enter` | Moves down one line. |
| `b` | Moves to the previous screen when supported. |
| `/word` | Searches for a word. |
| `n` | Finds the next match. |
| `q` | Exits the viewer. |




## head and tail Commands

### 1. head Command

### DEFINITION

The `head` command is used to display the first lines of a text file. By default, it shows the first 10 lines of the file.

For example, `head notes.txt` displays the first 10 lines of `notes.txt`.

Students use the `head` command to check the beginning of a file, read the first lines of an assignment, and quickly view information without opening the entire file.

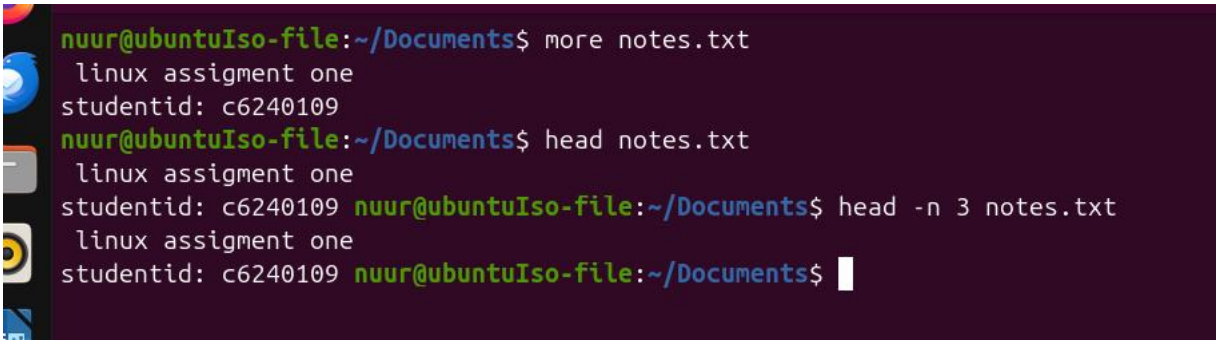
### SYNTAX

```
head [OPTION]... [FILE]...
```

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## OPTIONS AND SHORTCUTS

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-n` | Number of lines | Displays a specific number of lines. Example: `head -n 5 notes.txt` |
| `-c` | Number of bytes | Displays a specific number of bytes. Example: `head -c 20 notes.txt` |
| `-q` | Quiet mode | Does not display file names when reading multiple files. Example: `head -q file1.txt file2.txt` |
| `-v` | Verbose mode | Displays file names even when reading one file. Example: `head -v notes.txt` |
| `-n -5` | Exclude last lines | Displays all lines except the last five. Example: `head -n -5 notes.txt` |
| `--help` | Help | Displays command help. Example: `head --help` |




## . tail Command

## DEFINITION

The `tail` command is used to display the last lines of a text file. By default, it shows the last 10 lines.

For example, `tail notes.txt` displays the last 10 lines of `notes.txt`.

Students use the `tail` command to check the latest information in a file, read the last lines of documents, and monitor log files.

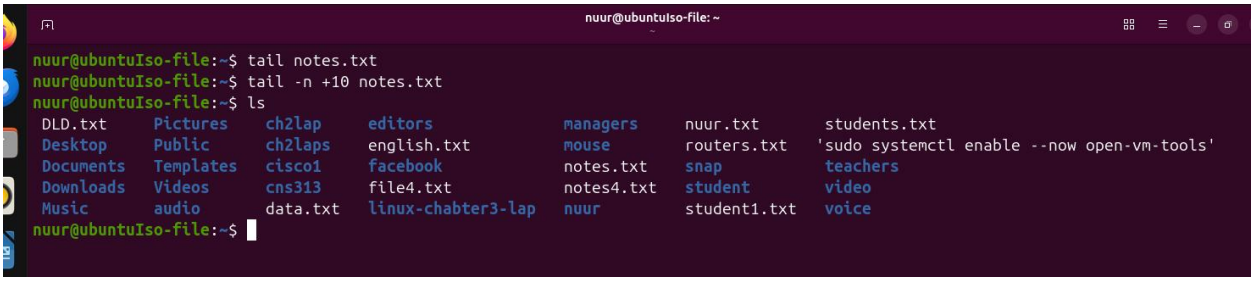
## SYNTAX

```
tail [OPTION]... [FILE]...
```

## OPTIONS AND SHORTCUTS

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|

| | | |
|---|---|---|
| `-n` | Number of lines | Displays a specific number of last lines. Example: `tail -n 5 notes.txt` |
| `-c` | Number of bytes | Displays the last number of bytes. Example: `tail -c 20 notes.txt` |
| `-f` | Follow | Continues displaying new lines added to a file. Example: `tail -f system.log` |
| `-F` | Follow by name | Keeps monitoring a file even if it is replaced or recreated. Example: `tail -F system.log` |
| `-q` | Quiet mode | Hides file names when reading multiple files. Example: `tail -q file1.txt file2.txt` |
| `-v` | Verbose mode | Displays file names in the output. Example: `tail -v notes.txt` |
| `-n +5` | Start from line | Displays the file starting from line five. Example: `tail -n +5 notes.txt` |
| `--help` | Help | Displays command help. Example: `tail --help` |




## find Command

### DEFINITION

The `find` command is used to search for files and directories in Linux. It searches for items based on their names, types, sizes, and modification times.

For example, `find . -name "notes.txt"` searches for a file named `notes.txt` in the current directory and its subdirectories.

Students use the `find` command to locate assignment files, search for folders, find files with specific extensions, and manage files more easily.

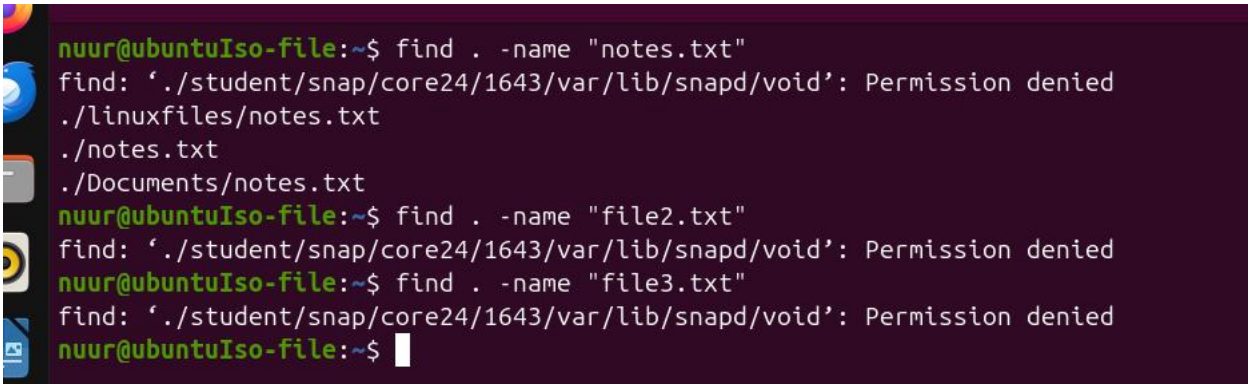
### SYNTAX

```
find [PATH] [EXPRESSION]
```

### OPTIONS AND SHORTCUTS

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|

| | | |
|---|---|---|
| `-name` | Search by name | Finds a file using its exact name pattern. Example: `find . -name "notes.txt"` |
| `-iname` | Ignore case | Searches for a name without considering uppercase or lowercase. Example: `find . -iname "notes.txt"` |
| `-type f` | Find files | Searches for regular files. Example: `find . -type f` |
| `-type d` | Find directories | Searches for directories. Example: `find . -type d` |
| `-size` | Search by size | Finds files based on their size. Example: `find . -size +1M` |
| `-empty` | Find empty items | Finds empty files and directories. Example: `find . -empty` |
| `-mtime` | Modification time | Finds files modified a certain number of days ago. Example: `find . -mtime -7` |
| `-delete` | Delete files | Deletes matching files. Example: `find . -name "*.tmp" -delete` |
| `-maxdepth` | Search depth | Limits how many directory levels are searched. Example: `find . -maxdepth 2` |
| `-user` | Search by owner | Finds files belonging to a user. Example: `find /home -user nuur` |




## `locate` Command

### DEFINITION

The `locate` command is used to quickly search for files and directories by name in a Linux system. It uses a database that stores file paths, so it can find files faster than the `find` command in many cases.

### SYNTAX

```
locate [OPTION]... PATTERN
```

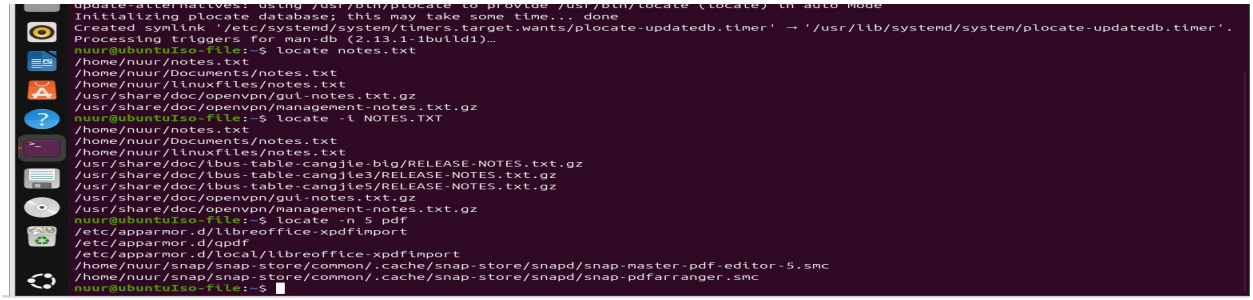
Example:

```
locate notes.txt
```

This command searches for paths containing the name `notes.txt`.

## OPTIONS AND SHORTCUTS

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-i` | Ignore Case | Searches without considering uppercase or lowercase letters. Example: `locate -i NOTES.txt` |
| `-n` | Limit Results | Displays only a specified number of results. Example: `locate -n 5 notes` |
| `-c` | Count Results | Displays the number of matching paths. Example: `locate -c notes.txt` |
| `-l` | Limit Output | Limits the number of results displayed. Example: `locate -l 10 notes` |
| `-b` | Basename | Matches the filename only, not the full directory path. Example: `locate -b '\notes.txt'` |
| `-r` | Regular Expression | Searches using a regular expression. Example: `locate -r 'notes.*txt'` |
| `-e` | Existing Files | Displays only files that currently exist. Example: `locate -e notes.txt` |
| `-S` | Database Statistics | Displays information about the locate database. Example: `locate -S` |
| `--help` | Help | Displays help information. Example: `locate --help` |
| `--version` | Version | Displays the program version. Example: `locate --version` |




# `grep` Command in Linux

## 1. Definition

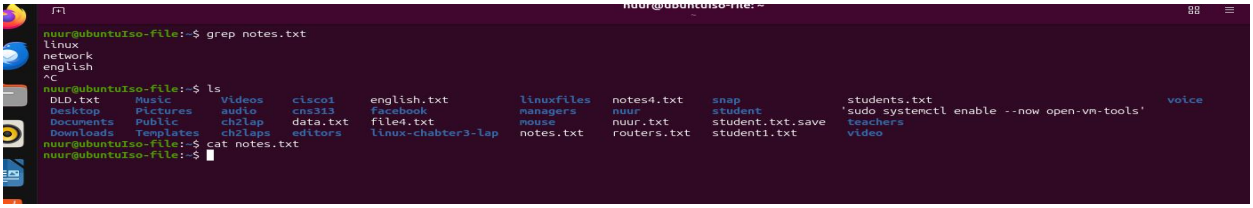
The `grep` command in Linux is used to search for specific words, letters, or patterns inside a file or text.

It helps users find the information they need without reading the entire file.

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## 4. Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-i` | Ignore Case | Searches without considering uppercase or lowercase. `grep -i "linux" file.txt` |
| `-n` | Line Number | Displays the line number where the word is found. `grep -n "linux" file.txt` |
| `-v` | Invert Match | Displays lines that do not contain the searched word. `grep -v "linux" file.txt` |
| `-r` | Recursive Search | Searches inside files in directories and subdirectories. `grep -r "linux" /home/nuur` |
| `-w` | Whole Word | Searches for an exact word. `grep -w "cat" file.txt` |
| `-c` | Count Matches | Counts the matching lines. `grep -c "linux" file.txt` |
| `-l` | List Files | Displays filenames containing the searched word. `grep -l "linux" *.txt` |
| `-o` | Only Matching | Displays only the matching text. `grep -o "linux" file.txt` |
| `-E` | Extended Regex | Allows searching with extended patterns. `grep -E "cat\|dog" file.txt` |
| `-A` | After Context | Displays matching lines and lines after them. `grep -A 2 "linux" file.txt` |
| `-B` | Before Context | Displays matching lines and lines before them. `grep -B 2 "linux" file.txt` |
| `-C` | Context | Displays matching lines with surrounding lines. `grep -C 2 "linux" file.txt` |




# `wc` Command in Linux

## 1. Definition

The `wc` command in Linux stands for Word Count. It is used to count the number of lines, words, characters, and bytes in a file.

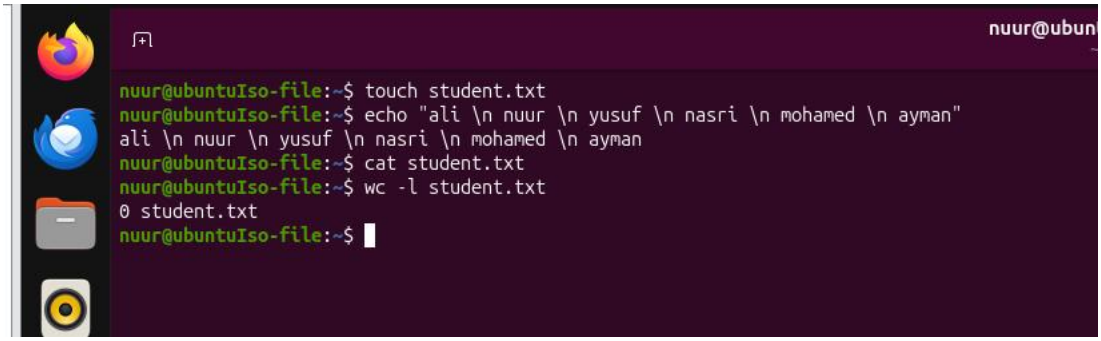
## 3. Syntax

```
wc [options] filename
```

## 4. Options and Shortcuts

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| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-l` | Lines | Counts the number of lines. `wc -l file.txt` |
| `-w` | Words | Counts the number of words. `wc -w file.txt` |
| `-c` | Bytes | Counts the number of bytes. `wc -c file.txt` |
| `-m` | Characters | Counts the number of characters. `wc -m file.txt` |
| `-L` | Maximum Line Length | Displays the length of the longest line. `wc -L file.txt` |




## `tr` Command in Linux

## The `tr` command in Linux stands for Translate. It is used to translate, replace, or delete specific characters from text.

```
tr [options] SET1 [SET2]
```

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-d` | Delete | Deletes selected characters. `tr -d 'a'` |
| `-s` | Squeeze | Replaces repeated characters with one character. `tr -s ' '` |
| `-c` | Complement | Selects characters not included in SET1. `tr -cd '0-9'` |
| `-t` | Truncate | Makes SET1 the same length as SET2. `tr -t 'abc' 'XY'` |




## `sort` Command in Linux

### 1. Definition

The `sort` command in Linux is used to arrange lines of text in a particular order. It can sort information alphabetically, numerically, or in reverse order.

### 3. Syntax

```
sort [options] filename
```

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## 4. Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-r` | Reverse | Sorts lines in reverse order. `sort -r file.txt` |
| `-n` | Numeric | Sorts numbers numerically. `sort -n numbers.txt` |
| `-u` | Unique | Removes duplicate lines while sorting. `sort -u file.txt` |
| `-f` | Ignore Case | Ignores uppercase and lowercase differences. `sort -f file.txt` |
| `-o` | Output File | Saves sorted results into a file. `sort file.txt -o sorted.txt` |
| `-k` | Key | Sorts using a selected field or column. `sort -k 2 file.txt` |
| `-t` | Separator | Specifies the field separator. `sort -t ':' -k 2 file.txt` |
| `-M` | Month | Sorts month names in calendar order. `sort -M months.txt` |
| `-R` | Random | Sorts lines in random order. `sort -R file.txt` |




# `cut` Command in Linux

## 1. Definition

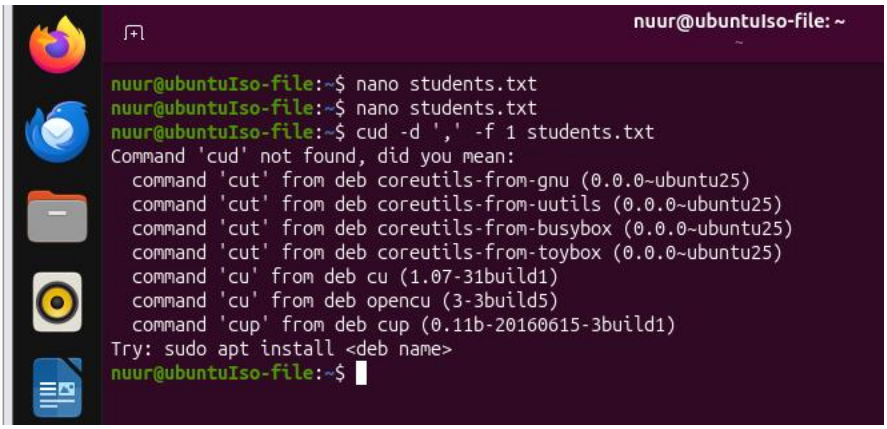
The `cut` command in Linux is used to extract specific parts of text from files or command output. It can select characters, bytes, or fields from each line.

It is useful when you want to display only the information you need instead of showing the entire line.

## 4. Options and Shortcuts

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| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-b` | Bytes | Extracts specific bytes. `cut -b 1-5 file.txt` |
| `-c` | Characters | Extracts specific characters. `cut -c 1-4 file.txt` |
| `-d` | Delimiter | Specifies the character used to separate fields. `cut -d ':' -f 1 file.txt` |
| `-f` | Fields | Selects specific fields or columns. `cut -f 2 file.txt` |
| `-s` | Suppress Lines | Does not display lines without the delimiter. `cut -d ',' -s -f 1 file.txt` |
| `--complement` | Complement | Displays everything except the selected fields or characters. `cut -d ',' -f 1 --complement file.txt` |
| `--output-delimiter` | Output Delimiter | Changes the separator in the output. `cut -d ',' -f 1,2 --output-delimiter='-' file.txt` |




## `paste` Command in Linux

### 1. Definition

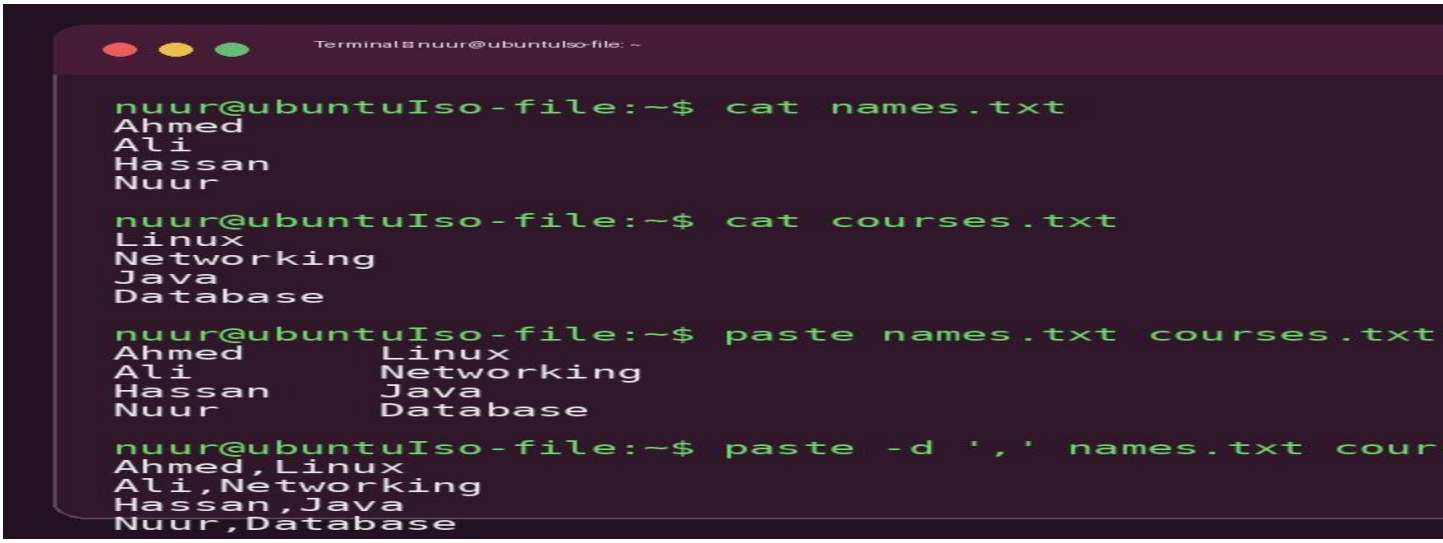
The `paste` command in Linux is used to combine lines from two or more files side by side. It joins the information from different files into one output.

By default, the `paste` command separates the combined information using a Tab character.

### 4. Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-d` | Delimiter | Changes the separator between columns. `paste -d ',' file1.txt file2.txt` |
| `-s` | Serial | Combines all lines of each file into one line. `paste -s file1.txt` |

| | | |
|---|---|---|
| `-z` | Zero Terminated | Uses a zero byte instead of a newline to separate input records. `paste -z file1.txt file2.txt` |
| `--help` | Help | Displays information about the command. `paste --help` |
| `--version` | Version | Displays the installed version. `paste --version` |




## `gzip`, `gunzip`, `bzip2`, and `bzcat` Commands in Linux

These four Linux commands are used to compress and decompress files. They help users reduce file sizes and access compressed information.

## 1. `gzip` Command in Linux

### 1. Definition

The `gzip` command stands for GNU Zip. It is used to compress files to reduce their size. It creates compressed files with the `.gz` extension.

For example, `students.txt` becomes `students.txt.gz`.

### 3. Syntax

```
gzip [options] filename
```

### 4. Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-k` | Keep Original | Keeps the original file after compression. `gzip -k file.txt` |
| `-d` | Decompress | Decompresses a `.gz` file. `gzip -d file.txt.gz` |
| `-r` | Recursive | Compresses files inside directories. `gzip -r folder/` |

| | | |
|---|---|---|
| `-v` | Verbose | Displays compression information. `gzip -v file.txt` |
| `-f` | Force | Forces compression even if the output file exists. `gzip -f file.txt` |
| `-1` | Fast Compression | Compresses faster with less compression. `gzip -1 file.txt` |
| `-9` | Best Compression | Uses maximum compression effort. `gzip -9 file.txt` |
| `-l` | List Information | Displays information about compressed files. `gzip -l file.txt.gz` |

## `gunzip` Command in Linux

### 1. Definition

The `gunzip` command is used to decompress files that have been compressed using gzip. It restores compressed `.gz` files to their original form.

For example, `students.txt.gz` becomes `students.txt`.

### 3. Syntax

```
gunzip [options] filename.gz
```

### 4. Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-k` | Keep Original | Keeps the compressed file after decompression. `gunzip -k file.gz` |
| `-c` | Standard Output | Displays decompressed content without saving a new file. `gunzip -c file.gz` |
| `-f` | Force | Forces decompression. `gunzip -f file.gz` |
| `-v` | Verbose | Displays decompression information. `gunzip -v file.gz` |
| `-r` | Recursive | Decompresses gzip files inside directories. `gunzip -r folder/` |
| `-t` | Test | Checks whether a compressed file is valid. `gunzip -t file.gz` |

## `bzip2` Command in Linux

### 1. Definition

The `bzip2` command is used to compress files and reduce their size. It creates compressed files with the `.bz2` extension.

It uses a different compression method from `gzip` and can sometimes produce smaller compressed files.

For example, `students.txt` becomes `students.txt.bz2`.

### 3. Syntax

```
bzip2 [options] filename
```

### 4. Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-k` | Keep Original | Keeps the original file after compression. `bzip2 -k file.txt` |
| `-d` | Decompress | Decompresses a `.bz2` file. `bzip2 -d file.txt.bz2` |
| `-c` | Standard Output | Sends compressed or decompressed data to the terminal output. `bzip2 -dc file.txt.bz2` |
| `-f` | Force | Forces compression or decompression. `bzip2 -f file.txt` |
| `-v` | Verbose | Displays compression information. `bzip2 -v file.txt` |
| `-t` | Test | Checks whether a compressed file is valid. `bzip2 -t file.txt.bz2` |
| `-1` | Fast Compression | Uses less compression effort and works faster. `bzip2 -1 file.txt` |
| `-9` | Best Compression | Uses maximum compression effort. `bzip2 -9 file.txt` |

## . `bzcat` Command in Linux

### 1. Definition

The `bzcat` command is used to display the contents of a file compressed with bzip2 without extracting the file permanently.

It reads `.bz2` files and displays their original text directly in the terminal.

### 3. Syntax

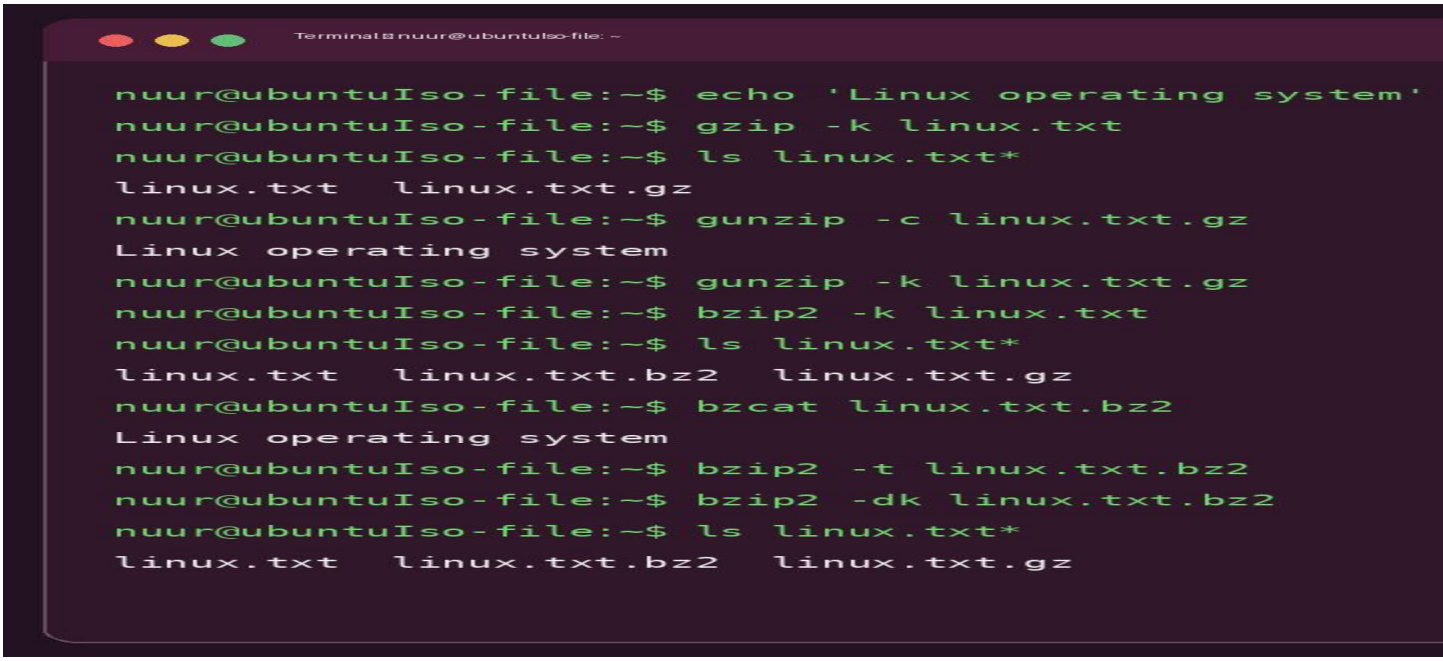
```
bzcat filename.bz2
```

### 4. Options and Shortcuts

`bzcat` is mainly used to display decompressed content. It has fewer commonly used options than `bzip2`.

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|

| | | |
|---|---|---|
| `-s` | Small Memory | Uses less memory when decompressing. `bzcat -s file.txt.bz2` |
| `--help` | Help | Displays available command information. `bzcat --help` |
| `--version` | Version | Displays version information. `bzcat --version` |




## `nano` Command in Linux

### 1. Definition

The `nano` command in Linux is a simple text editor used to create, open, edit, and save text files directly from the terminal.

It is easy to use, especially for beginners, because it displays its keyboard shortcuts at the bottom of the screen.

For example, you can use `nano` to create a text file, write information, edit existing content, and save your work.

### 4. Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-B` | Backup | Creates a backup before saving changes. `nano -B file.txt` |
| `-l` | Line Numbers | Displays line numbers. `nano -l file.txt` |
| `-m` | Mouse Support | Enables mouse support. `nano -m file.txt` |

| | | |
|---|---|---|
| `-v` | View Mode | Opens a file in read-only mode. `nano -v file.txt` |
| `-i` | Auto Indent | Automatically indents new lines. `nano -i file.txt` |
| `-c` | Constant Cursor | Displays cursor position information. `nano -c file.txt` |
| `-w` | No Wrap | Prevents long lines from being automatically wrapped. `nano -w file.txt` |
| `Ctrl + O` | Write Out | Saves the current file. |
| `Ctrl + X` | Exit | Exits the Nano editor. |
| `Ctrl + W` | Search | Searches for a word or phrase. |
| `Ctrl + K` | Cut Line | Cuts the current line. |
| `Ctrl + U` | Paste | Pastes previously cut text. |
| `Ctrl + G` | Help | Displays Nano help information. |
| `Ctrl + C` | Cursor Position | Displays the current cursor position. |
| `Ctrl + \` | Replace | Finds and replaces text. |
| `Alt + U` | Undo | Reverses the last editing action. |
| `Alt + E` | Redo | Restores an undone editing action. |

Note: In Nano, `^` means Ctrl and `M-` means Alt in the shortcut display.

## `ps` Command in Linux

### 1. Definition

The `ps` command in Linux stands for Process Status. It is used to display information about processes currently running on the system.

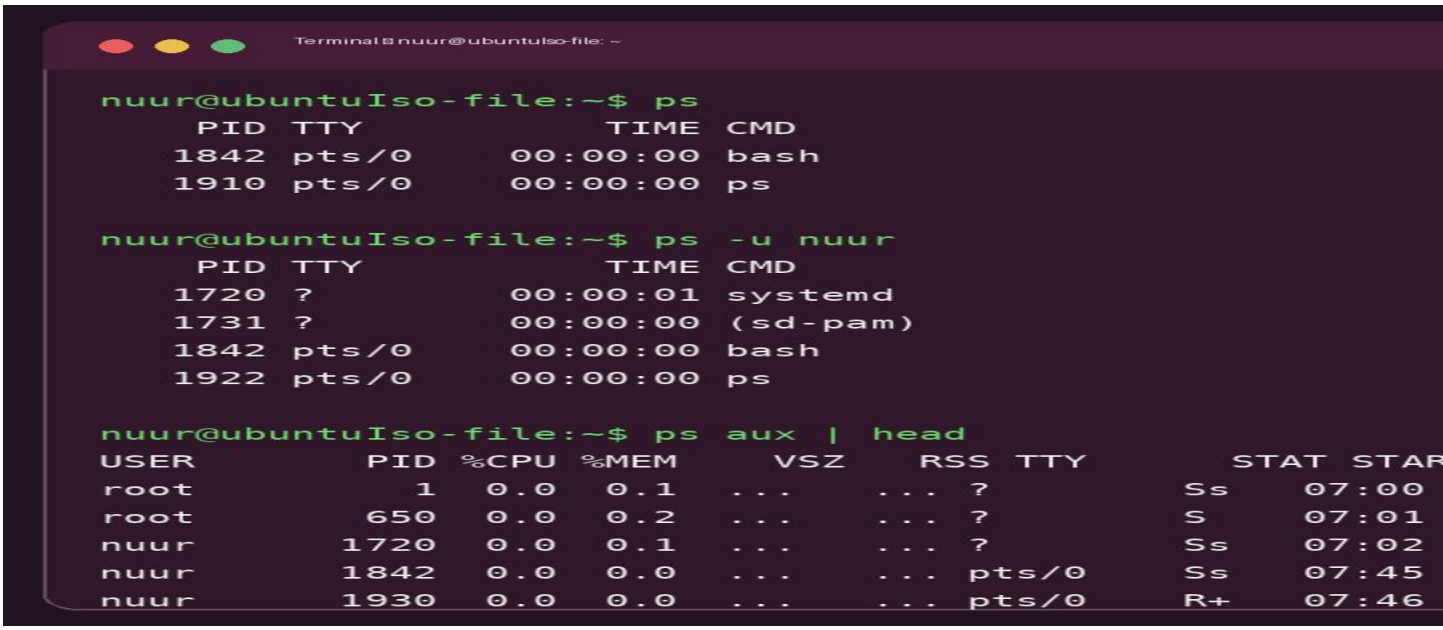
A process is a program or application that is currently running on the computer.

The `ps` command helps users check running programs, process IDs (PIDs), and the status of processes.

### 4. Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-e` | Every Process | Displays all running processes. `ps -e` |
| `-f` | Full Format | Displays detailed process information. `ps -f` |
| `-u` | User Format | Displays processes belonging to a specific user. `ps -u nuur` |
| `-a` | All Users | Displays processes associated with terminals for all users. `ps -a` |

| | | |
|---|---|---|
| `-x` | No Terminal | Displays processes without a controlling terminal as well. `ps -x` |
| `-aux` | Detailed Process List | Common BSD-style option to display processes from all users. `ps aux` |
| `-l` | Long Format | Displays additional process details. `ps -l` |
| `-p` | Process ID | Displays information about a specific process. `ps -p 1234` |
| `-C` | Command Name | Displays processes matching a command name. `ps -C firefox` |
| `--sort` | Sort Processes | Sorts process information by a selected field. `ps aux --sort=-%mem` |
| `-H` | Process Hierarchy | Displays processes in a hierarchical format. `ps -eH` |




## `top` Command in Linux

### 1. Definition

The `top` command in Linux is used to monitor running processes and system performance in real time.

It displays information about CPU usage, memory usage, running processes, and system uptime.

Unlike the `ps` command, which displays a snapshot of processes, the `top` command continuously updates the information while it is running.

### 4. Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-b` | Batch Mode | Runs top without interactive controls. `top -b` |
| `-d` | Delay | Sets the refresh interval in seconds. `top -d 2` |
| `-n` | Number of Iterations | Sets how many updates to display. `top -n 5` |
| `-u` | User Filter | Displays processes belonging to a specific user. `top -u nuur` |
| `-p` | Process ID | Monitors a specific process. `top -p 1234` |
| `-H` | Threads | Displays individual threads. `top -H` |
| `-c` | Command Line | Displays full command lines instead of command names. `top -c` |
| `-i` | Idle Processes | Starts with idle and zombie processes hidden. `top -i` |
| `-o` | Sort Field | Sorts processes by a selected field. `top -o %MEM` |
| `-w` | Width | Sets the output width. `top -w 120` |

### Important keyboard shortcuts inside top

| Shortcut | Name | Description |
|---|---|---|
| `P` | CPU Sort | Sorts processes by CPU usage. |
| `M` | Memory Sort | Sorts processes by memory usage. |
| `T` | Time Sort | Sorts processes by CPU time. |
| `k` | Kill Process | Sends a signal to a selected process. |
| `r` | Renice | Changes the priority of a process. |
| `1` | CPU Cores | Displays individual CPU core usage. |
| `c` | Command | Toggles between command names and full commands. |
| `h` | Help | Displays help information. |
| `q` | Quit | Exits the top command. |

## `sleep` Command in Linux

### 1. Definition

The `sleep` command in Linux is used to pause or delay the execution of a command or program for a specified amount of time.

It makes the terminal or script wait before continuing to the next command.

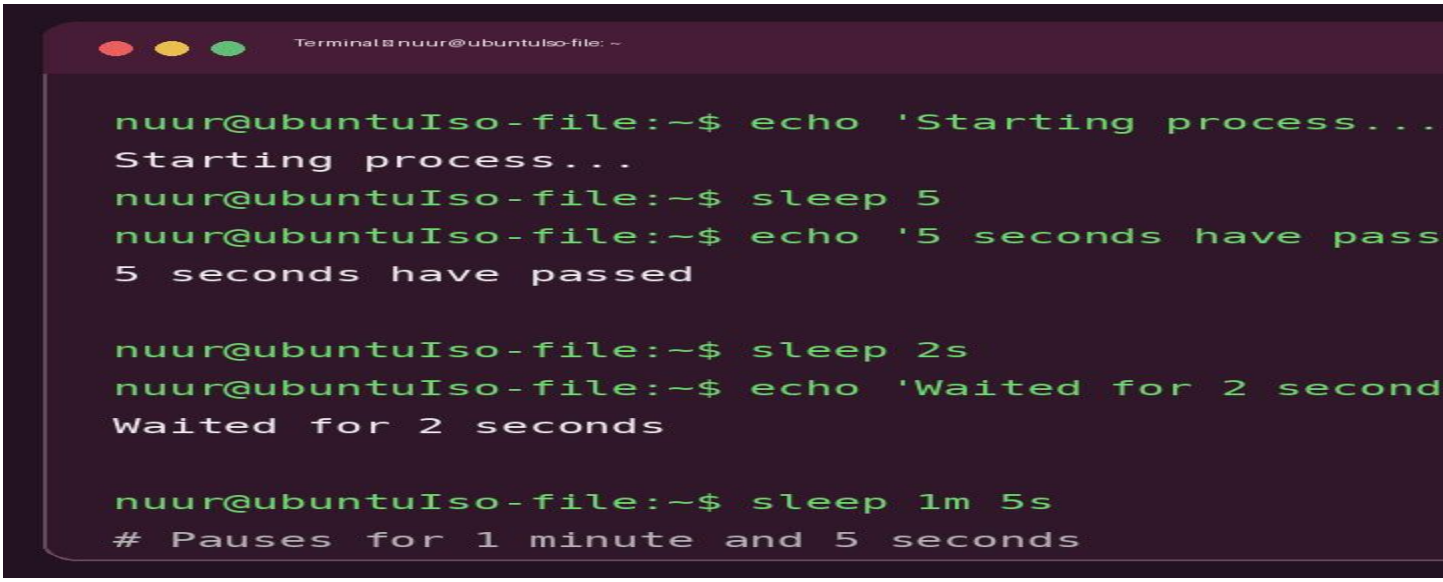
For example, `sleep 5` pauses execution for five seconds.

### 4. Options and Shortcuts

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The GNU/Linux `sleep` command has very few options. Most of its usage depends on specifying the time value.

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `--help` | Help | Displays information about the command. `sleep --help` |
| `--version` | Version | Displays the installed version. `sleep --version` |
| `NUMBER` | Seconds | Pauses execution for the specified seconds. `sleep 5` |
| `m` | Minutes | Pauses execution for minutes. `sleep 2m` |
| `h` | Hours | Pauses execution for hours. `sleep 1h` |
| `d` | Days | Pauses execution for days. `sleep 1d` |




## `jobs` Command in Linux

### 1. Definition

The `jobs` command in Linux is used to display jobs that are currently running or stopped in the background of the current terminal session.

It helps users manage multiple tasks without closing the terminal.

The `jobs` command is commonly used together with background processes, `&`, and keyboard shortcuts such as `Ctrl + Z`.

### 4. Options and Shortcuts

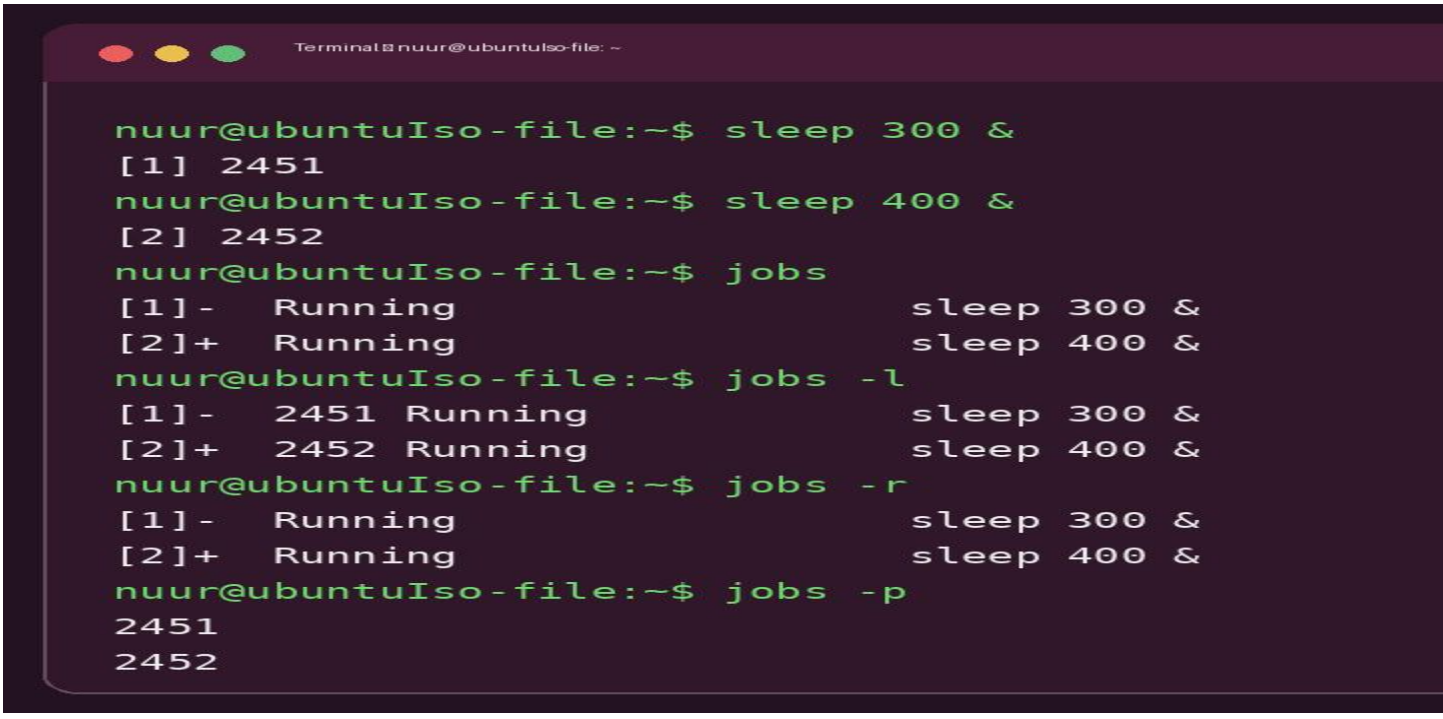
| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|

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| | | |
|---|---|---|
| `-l` | Long Format | Displays job numbers and process IDs. `jobs -l` |
| `-n` | New Jobs | Displays jobs whose status has changed since the last notification. `jobs -n` |
| `-p` | Process ID | Displays process IDs of job leaders. `jobs -p` |
| `-r` | Running Jobs | Displays only running jobs. `jobs -r` |
| `-s` | Stopped Jobs | Displays only stopped jobs. `jobs -s` |
| `-x` | Execute | Replaces job specifications with corresponding process IDs before running a command. `jobs -x echo %1` |

Important keyboard shortcuts and symbols

| Shortcut / Symbol | Name | Description |
|---|---|---|
| `&` | Background | Starts a command in the background. `sleep 100 &` |
| `Ctrl + Z` | Suspend | Temporarily stops a foreground job. |
| `fg` | Foreground | Brings a background or stopped job to the foreground. |
| `bg` | Background Resume | Continues a stopped job in the background. |
| `%1` | Job Number | Refers to job number 1. |




## `kill` Command in Linux

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## 1. Definition

The `kill` command in Linux is used to send signals to running processes. It is commonly used to stop or terminate a program that is running on the computer.

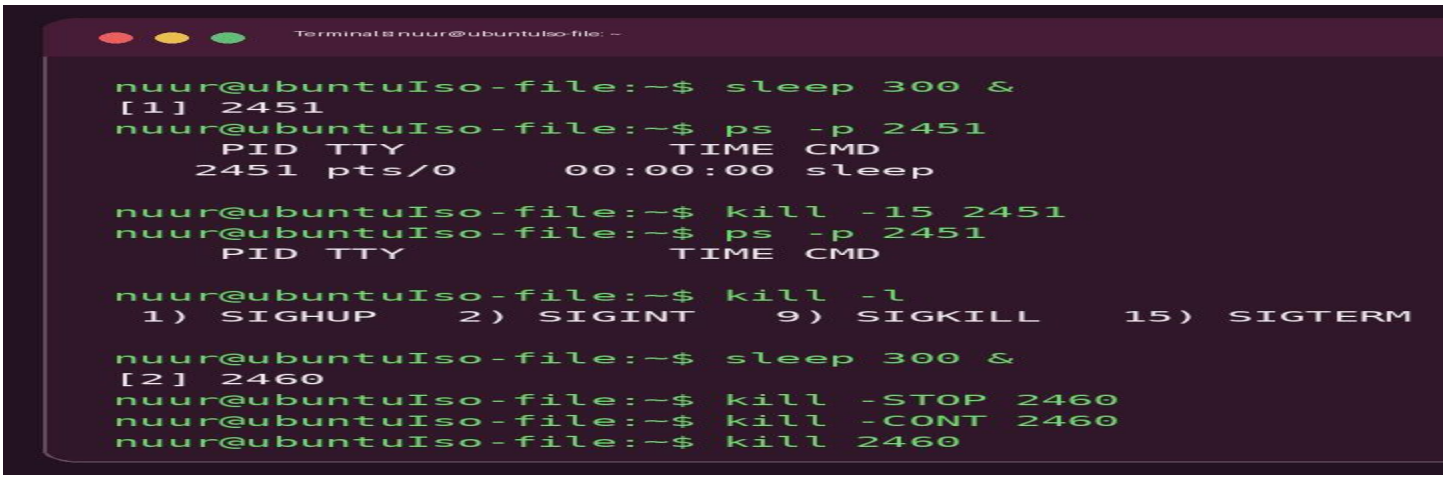
Every running process has a unique Process ID (PID), which can be used with the `kill` command to control that process.

For example, `kill 1234` sends a termination signal to the process with PID 1234.

## 4. Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-l` | List Signals | Displays available signal names. `kill -l` |
| `-15` | SIGTERM | Requests a process to terminate normally. `kill -15 1234` |
| `-9` | SIGKILL | Forcefully terminates a process. `kill -9 1234` |
| `-2` | SIGINT | Interrupts a process. `kill -2 1234` |
| `-1` | SIGHUP | Sends a hangup signal, often used to make services reload configuration. `kill -1 1234` |
| `-STOP` | Stop Process | Stops a process without terminating it. `kill -STOP 1234` |
| `-CONT` | Continue Process | Resumes a stopped process. `kill -CONT 1234` |
| `-0` | Check Process | Checks whether a process exists and whether you have permission to signal it. `kill -0 1234` |

Important: The `kill` command does not always terminate a process. It sends a signal, and the result depends on the signal used




## `nice` Command in Linux

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## 1. Definition

The `nice` command in Linux is used to run a program with a specific priority level. It controls how much CPU priority a process receives compared to other running processes.

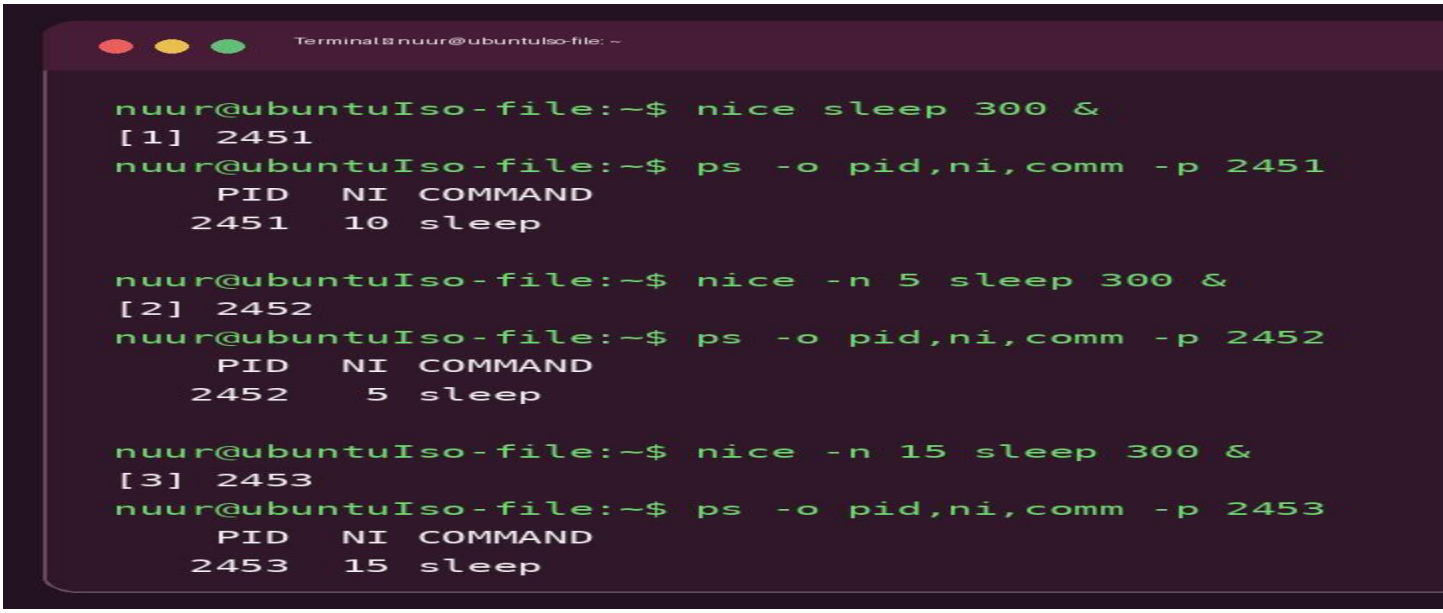
Linux uses a value called niceness to determine process priority.

The niceness values range from -20 to 19:

A higher niceness value means the process is more willing to give CPU time to other processes.

## 3. Syntax

```
nice [options] command
```




# `uname -r` Command in Linux

## 1. Definition

The `uname -r` command in Linux is used to display the kernel release version of the operating system.

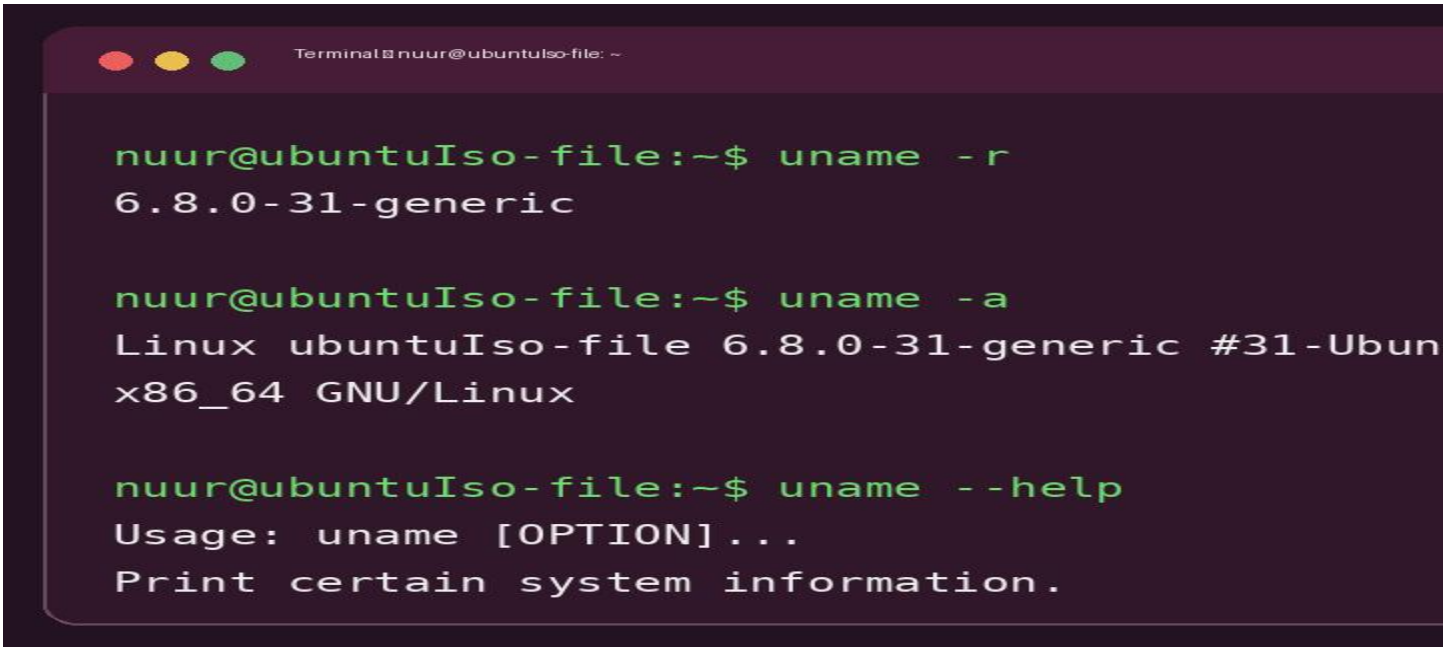
The kernel is the main part of Linux that manages communication between computer hardware and software.

The `-r` option stands for Kernel Release. It displays the version of the Linux kernel currently running on the computer.

## Options and Shortcuts

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| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-r` | Kernel Release | Displays the kernel release version. `uname -r` |
| `-a` | All Information | Displays all available system information. `uname -a` |
| `-s` | Kernel Name | Displays the kernel name. `uname -s` |
| `-n` | Network Node | Displays the computer hostname. `uname -n` |
| `-v` | Kernel Version | Displays the kernel version details. `uname -v` |
| `-m` | Machine Hardware | Displays the machine hardware architecture. `uname -m` |
| `-p` | Processor | Displays the processor type, if available. `uname -p` |
| `-i` | Hardware Platform | Displays the hardware platform, if available. `uname -i` |
| `-o` | Operating System | Displays the operating system name. `uname -o` |
| `--help` | Help | Displays command information. `uname --help` |
| `--version` | Version | Displays the uname utility version. `uname --version` |




## `hostnamectl` Command in Linux

### 1. Definition

The `hostnamectl` command in Linux is used to display and manage the hostname of a computer.

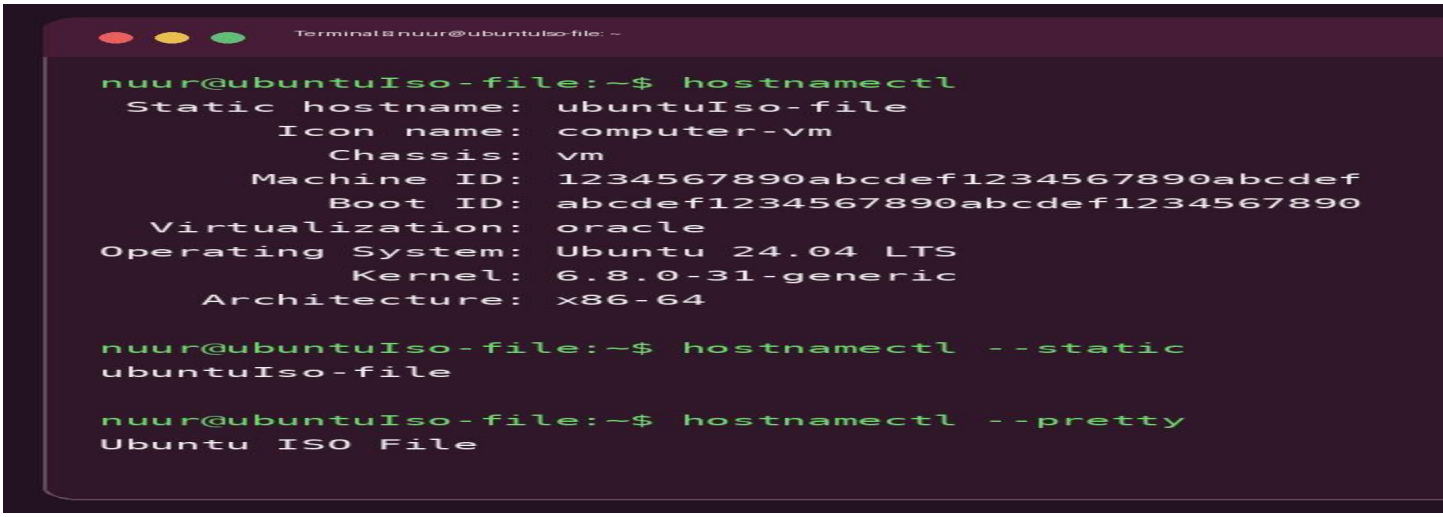
A hostname is the name assigned to a computer on a network. It helps identify the computer when communicating with other devices.

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The `hostnamectl` command is part of the systemd system management tool. It allows users to view and change the computer name, static hostname, pretty hostname, and other system information.

## Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `status` | Status | Displays hostname and system information. `hostnamectl status` |
| `--static` | Static Hostname | Displays the static hostname. `hostnamectl --static` |
| `--transient` | Transient Hostname | Displays the temporary hostname. `hostnamectl --transient` |
| `--pretty` | Pretty Hostname | Displays the descriptive hostname. `hostnamectl --pretty` |
| `set-hostname` | Change Hostname | Changes the computer hostname. `sudo hostnamectl set-hostname nuur-pc` |
| `set-icon-name` | Icon Name | Changes the system icon name. `sudo hostnamectl set-icon-name computer` |
| `set-chassis` | Chassis Type | Sets the computer type. `sudo hostnamectl set-chassis desktop` |
| `set-deployment` | Deployment | Sets the deployment environment. `sudo hostnamectl set-deployment production` |
| `set-location` | Location | Sets the system location description. `sudo hostnamectl set-location office` |
| `--help` | Help | Displays command information. `hostnamectl --help` |
| `--version` | Version | Displays the installed version. `hostnamectl --version` |



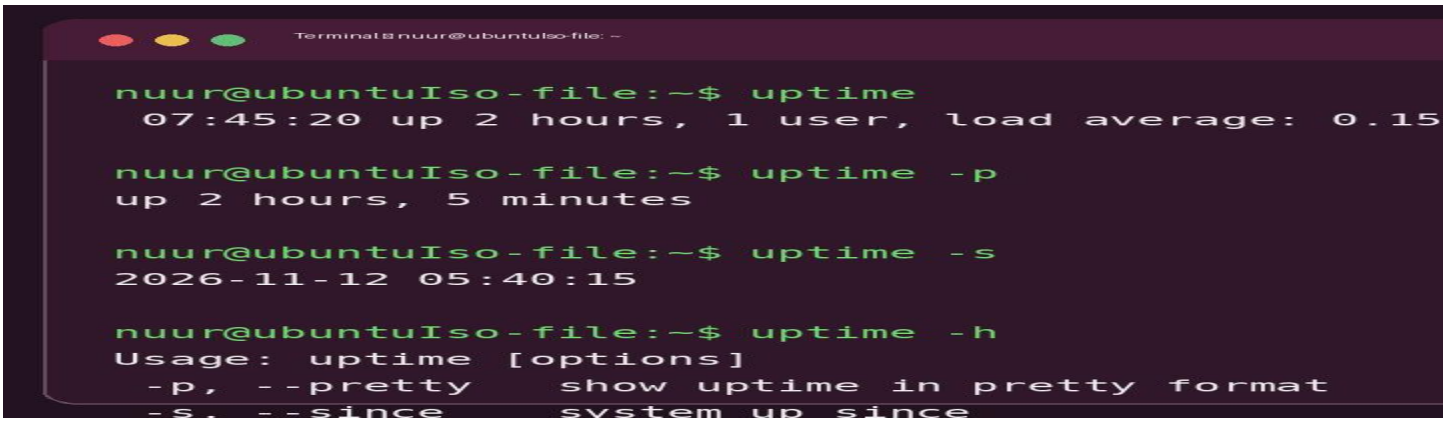

## `uptime` Command in Linux

### 1. Definition

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The `uptime` command in Linux is used to display how long the computer has been running since it was last started or rebooted.

It also displays the current system time, number of logged-in users, and system load averages.




## `free` Command in Linux

### 1. Definition

The `free` command in Linux is used to display information about the computer's memory (RAM) and swap memory.

It shows how much memory is available, how much is being used, and how much is free.

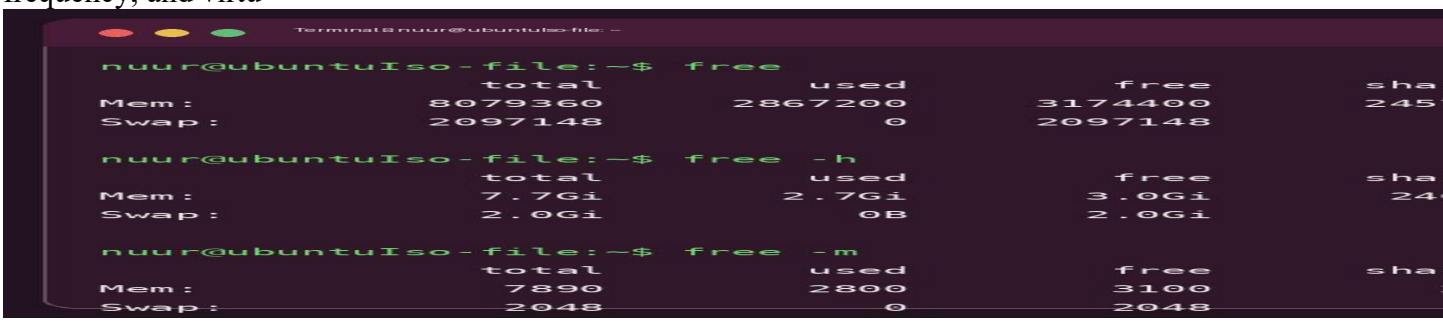
The command helps users understand memory usage and monitor system performance.

## 4`lscpu` Command in Linux

### 1. Definition

The `lscpu` command in Linux is used to display information about the computer's CPU (Central Processing Unit).

It provides details about the processor architecture, CPU model, number of cores, threads, CPU frequency, and virtu



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## . Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-h` | Human Readable | Displays memory in KB, MB, or GB. `free -h` |
| `-b` | Bytes | Displays memory in bytes. `free -b` |
| `-k` | Kilobytes | Displays memory in kilobytes. `free -k` |
| `-m` | Megabytes | Displays memory in megabytes. `free -m` |
| `-g` | Gigabytes | Displays memory in gigabytes. `free -g` |
| `-t` | Total | Displays total RAM and swap memory. `free -t` |
| `-s` | Seconds | Continuously displays memory information at a specified interval. `free -s 2` |
| `-c` | Count | Sets the number of updates to display. `free -s 2 -c 5` |
| `-w` | Wide Output | Displays memory information in a wider format. `free -w` |
| `-l` | Low and High | Displays low and high memory statistics. `free -l` |

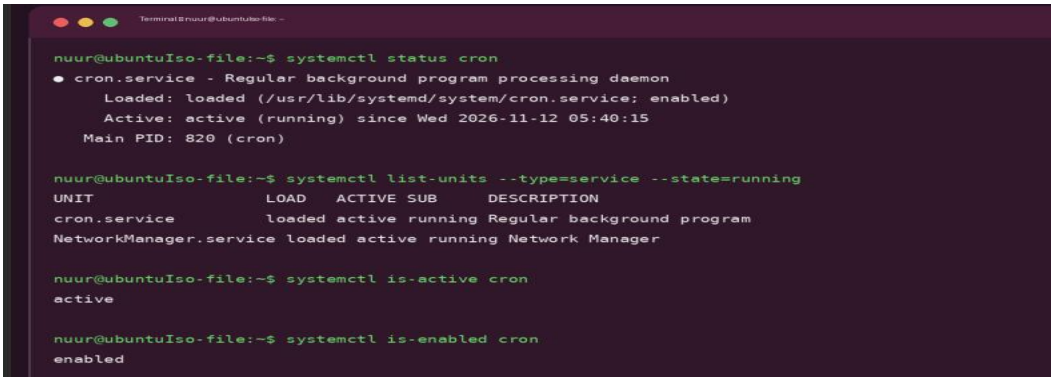
# `systemctl` Command in Linux

## 1. Definition

The `systemctl` command in Linux is used to manage and control system services and the systemd system manager.

A service is a program that runs in the background to perform specific tasks, such as managing network connections, scheduling tasks, and controlling system activities.

The `systemctl` command allows users to start, stop, restart, enable, disable, and check the status of services.




# `journalctl` Command in Linux

## 1. Definition

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The `journalctl` command in Linux is used to display and manage system logs collected by the systemd journal.

System logs are records of activities that happen on a Linux computer. They contain information about system startup, errors, services, applications, and other system events.

The `journalctl` command helps users read these records and troubleshoot system problems.

## Options and Shortcuts

| Option / Shortcut | Name | Description & Example Usage |
|---|---|---|
| `-n` | Number of Lines | Displays the latest log entries. `journalctl -n 20` |
| `-f` | Follow | Continuously displays new log entries. `journalctl -f` |
| `-b` | Current Boot | Displays logs from the current boot. `journalctl -b` |
| `-k` | Kernel Logs | Displays kernel messages. `journalctl -k` |
| `-u` | Unit | Displays logs for a specific service. `journalctl -u cron` |
| `-p` | Priority | Filters logs by priority level. `journalctl -p err` |
| `-r` | Reverse | Displays the newest logs first. `journalctl -r` |
| `-e` | End | Jumps to the end of the log. `journalctl -e` |
| `--since` | Start Time | Displays logs from a specified time. `journalctl --since today` |
| `--until` | End Time | Displays logs up to a specified time. `journalctl --until today` |
| `--no-pager` | No Pager | Displays output directly in the terminal. `journalctl --no-pager` |
| `--disk-usage` | Disk Usage | Displays disk space used by journal logs. `journalctl --disk-usage` |
| `--list-boots` | Boot History | Displays recorded system boots. `journalctl --list-boots` |
| `--help` | Help | Displays available options. `journalctl --help` |

Thank you for my self

End of the chepters